YFG FCL ANTISTATIC CHAMBER.

Operator / Maintenance Manual

Manufactured by:
Paveway Automation Pvt. Ltd.
Sanghvi Steel Complex,
128/2, TELCO Road,
Chinchwad,
Pune - 411019

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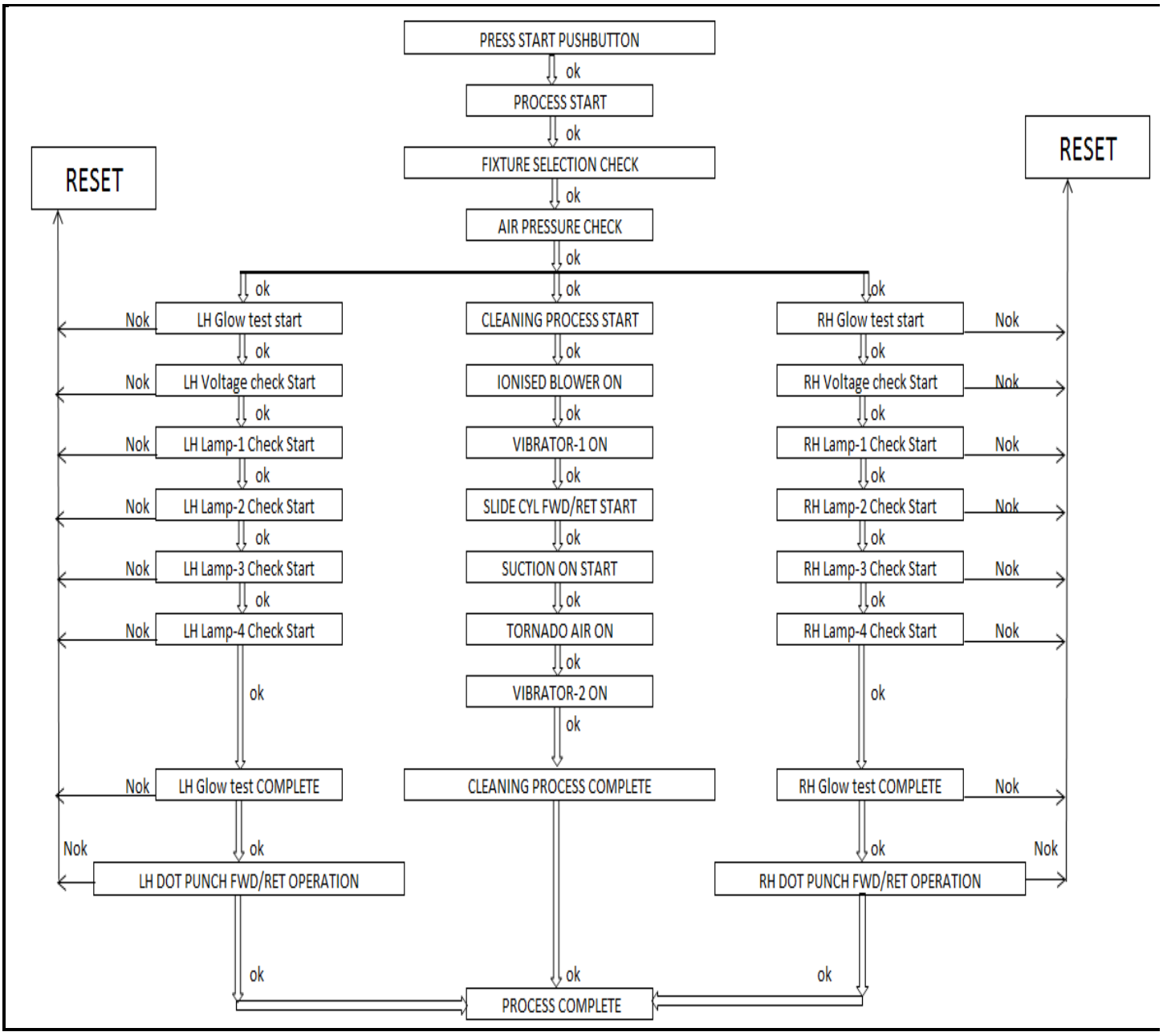
SYSTEM INFORMATION:

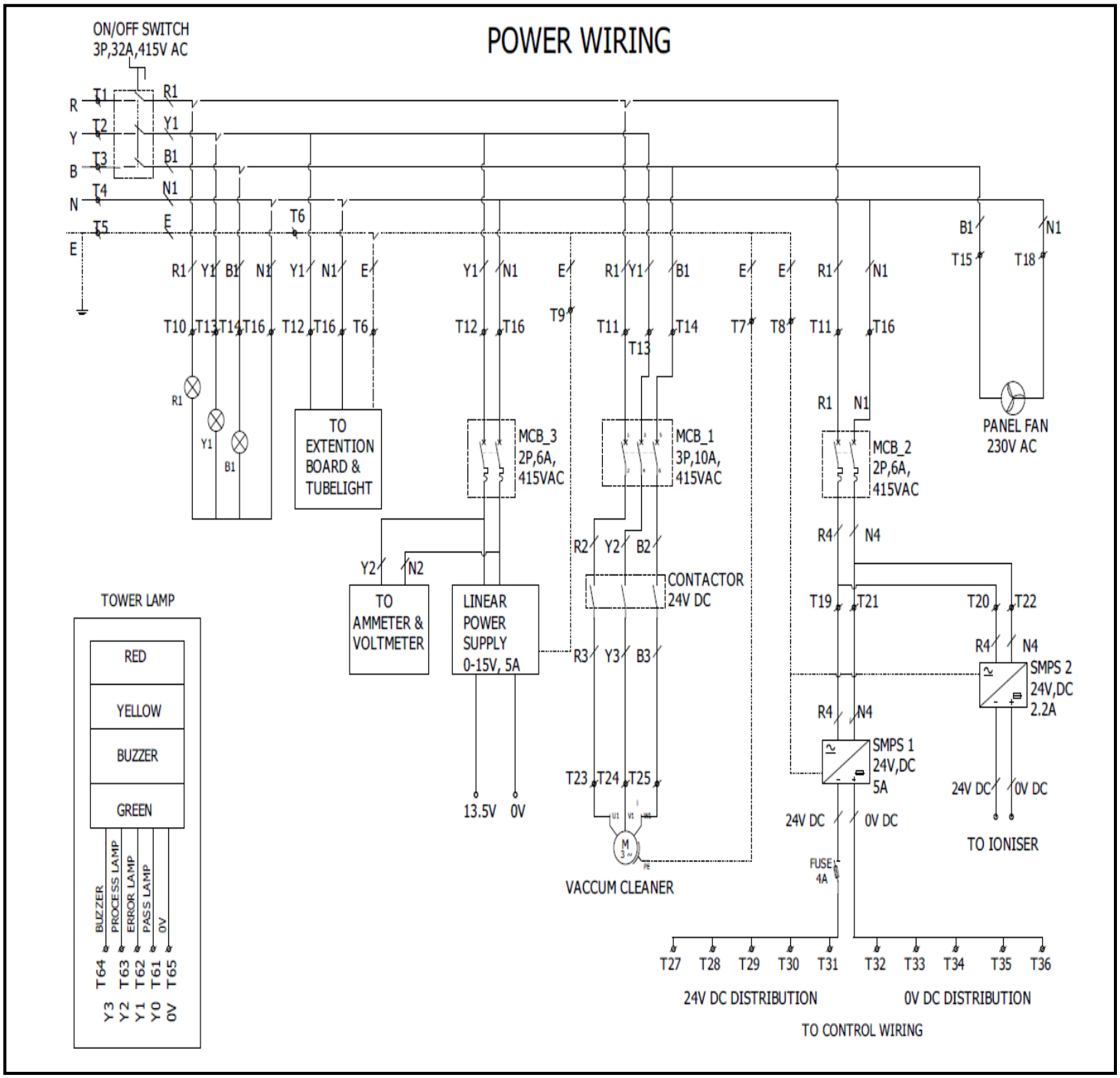
Name :	YFG FCL ANTISTATIC CHAMBER.
Customer :	Marelli Motherson Automotive Lighting India Pvt Ltd. Gat No.-165/1,165/2 &165/3 Village Ambethan, Dwarka School, Taluka Khed, Chakan, Pune-410501
Manufacturing Date :	November, 2021.
Supply Voltage :	415VAC, 50/60 Hz
Power Consumption:	1.6kw
Manufacturer :	**Paveway Automation Pvt. Ltd.** Sanghvi Steel Complex, 128/2, Telco Road, Chinchwad, Pune- 411019. tel: 020-27450840
Email ID :	***applications@pavewayautomation.com***

PROCESS SEQUENCE:

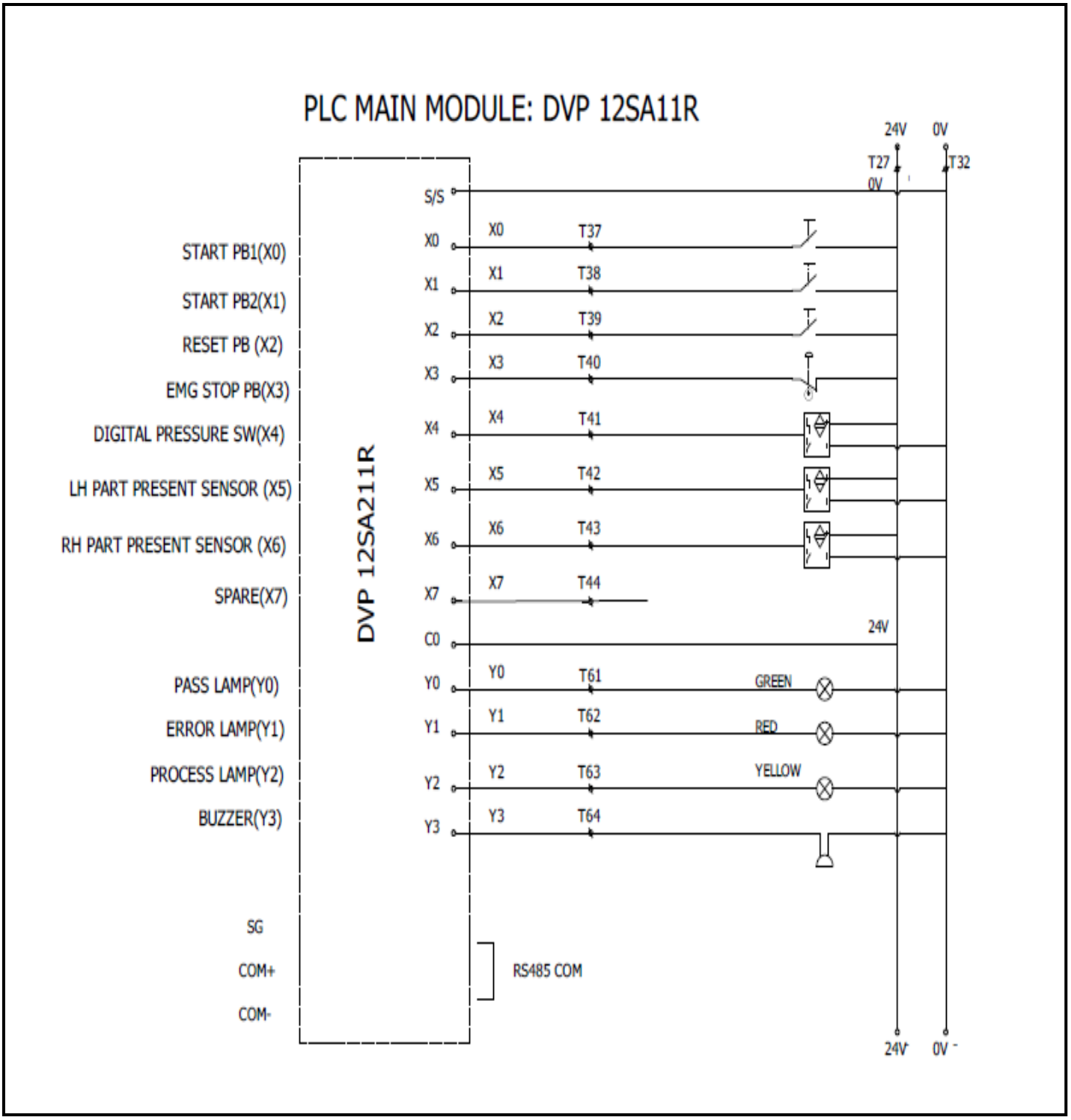


WIRING DIAGRAMS:

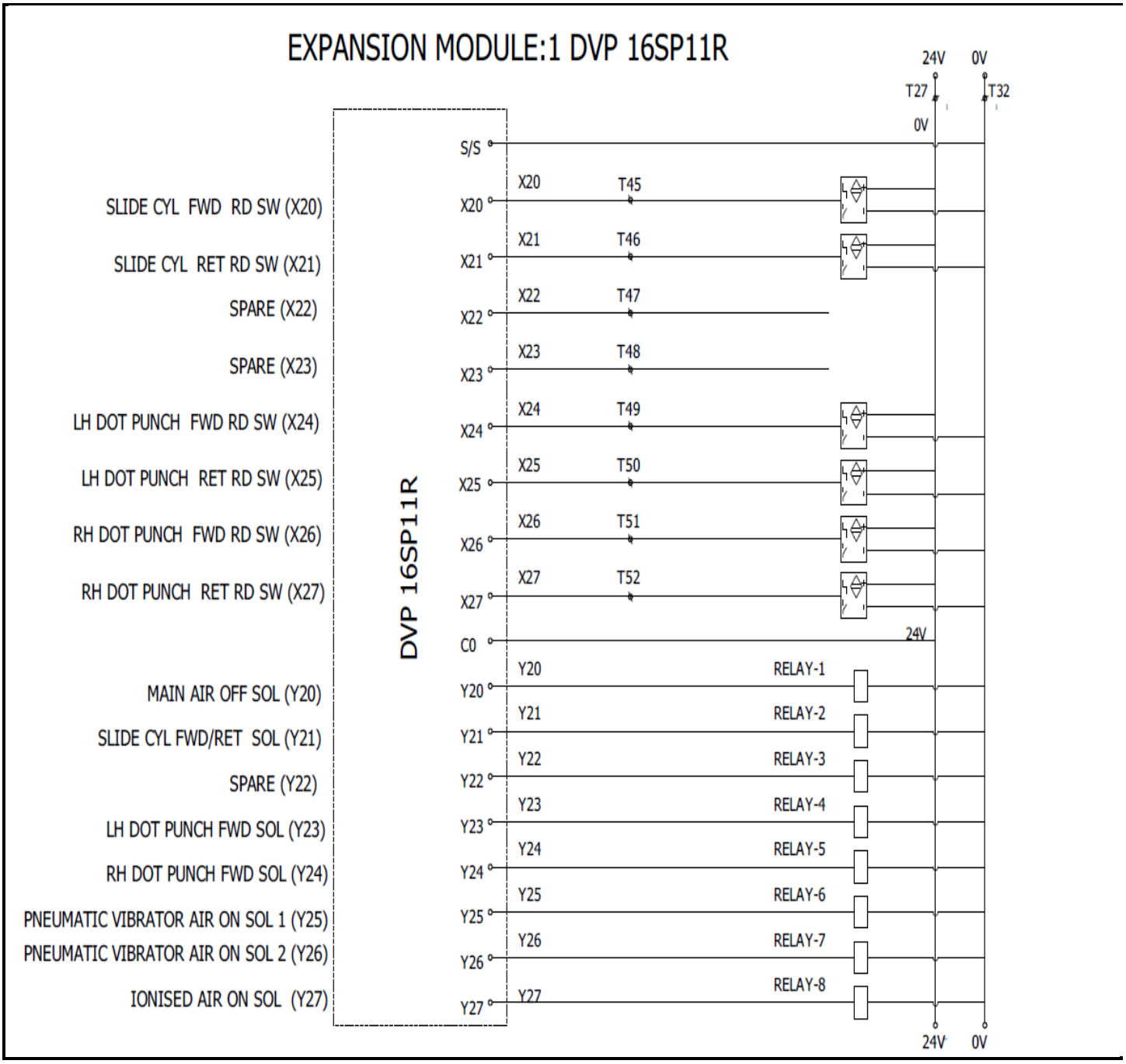
POWER DIAGRAM



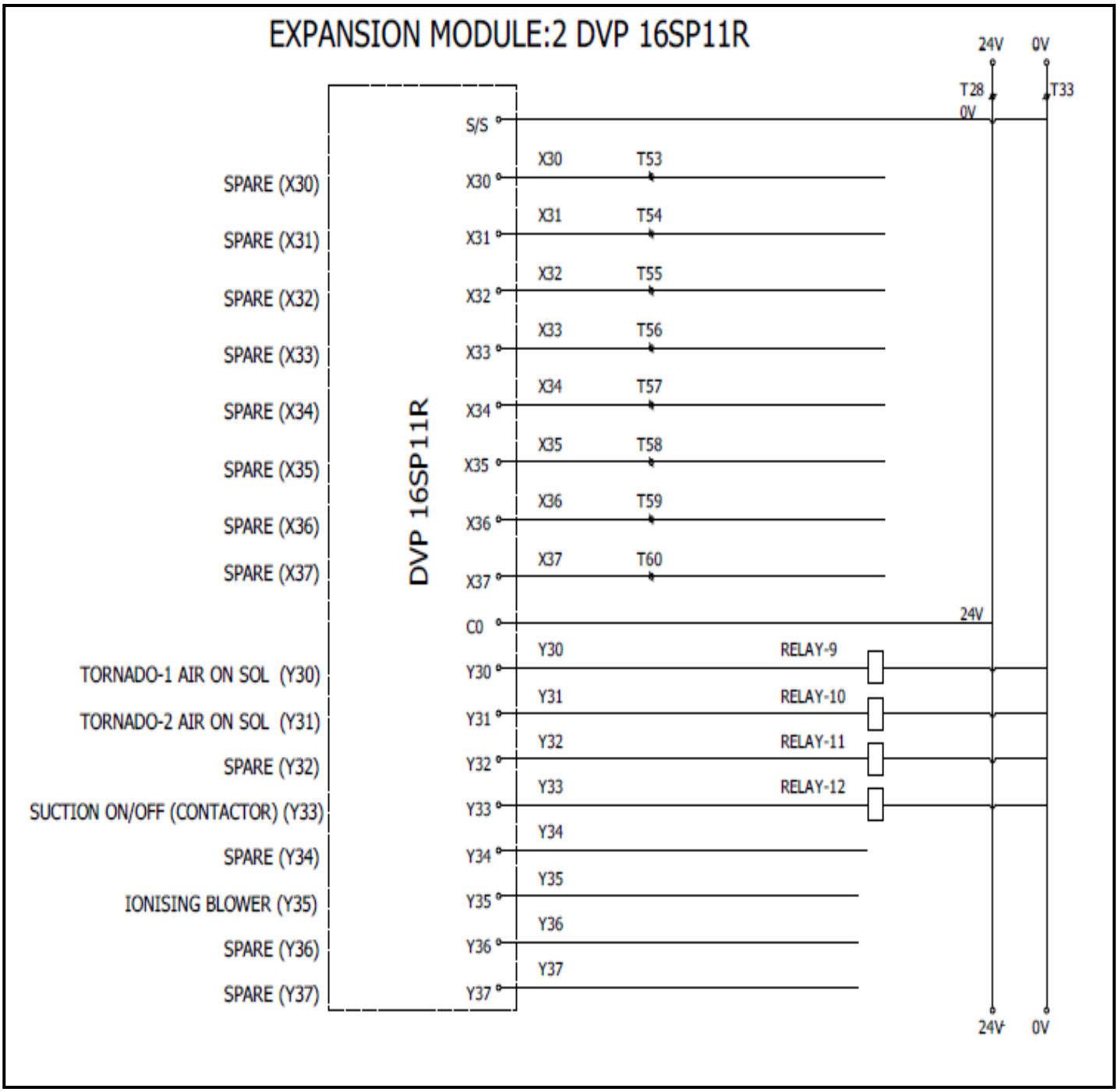
MAIN PLC MODULE-DVP12SA11R



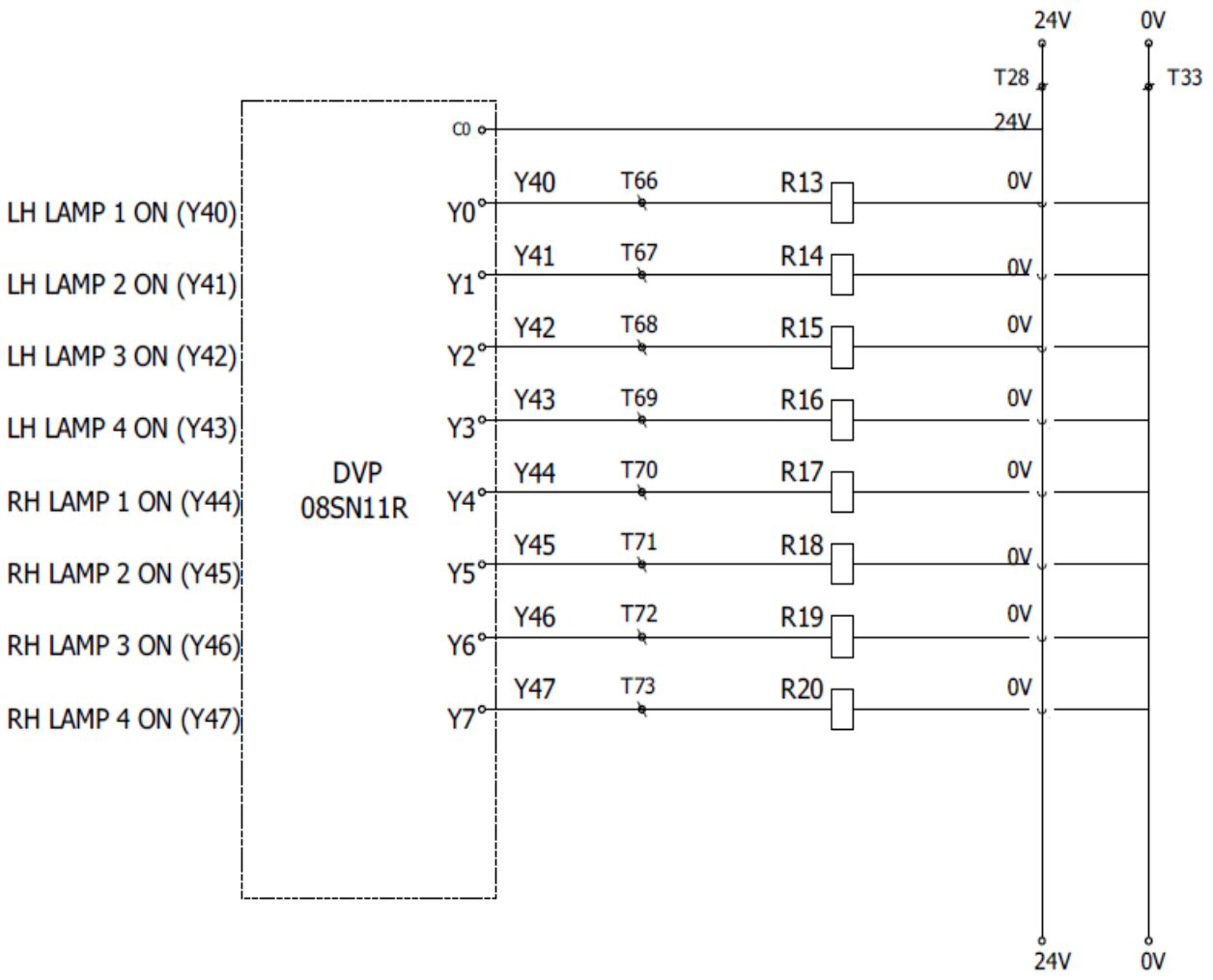
EXPANSION MODULE 1: DVP 16SP11R-1 CONTROL DIAGRAM



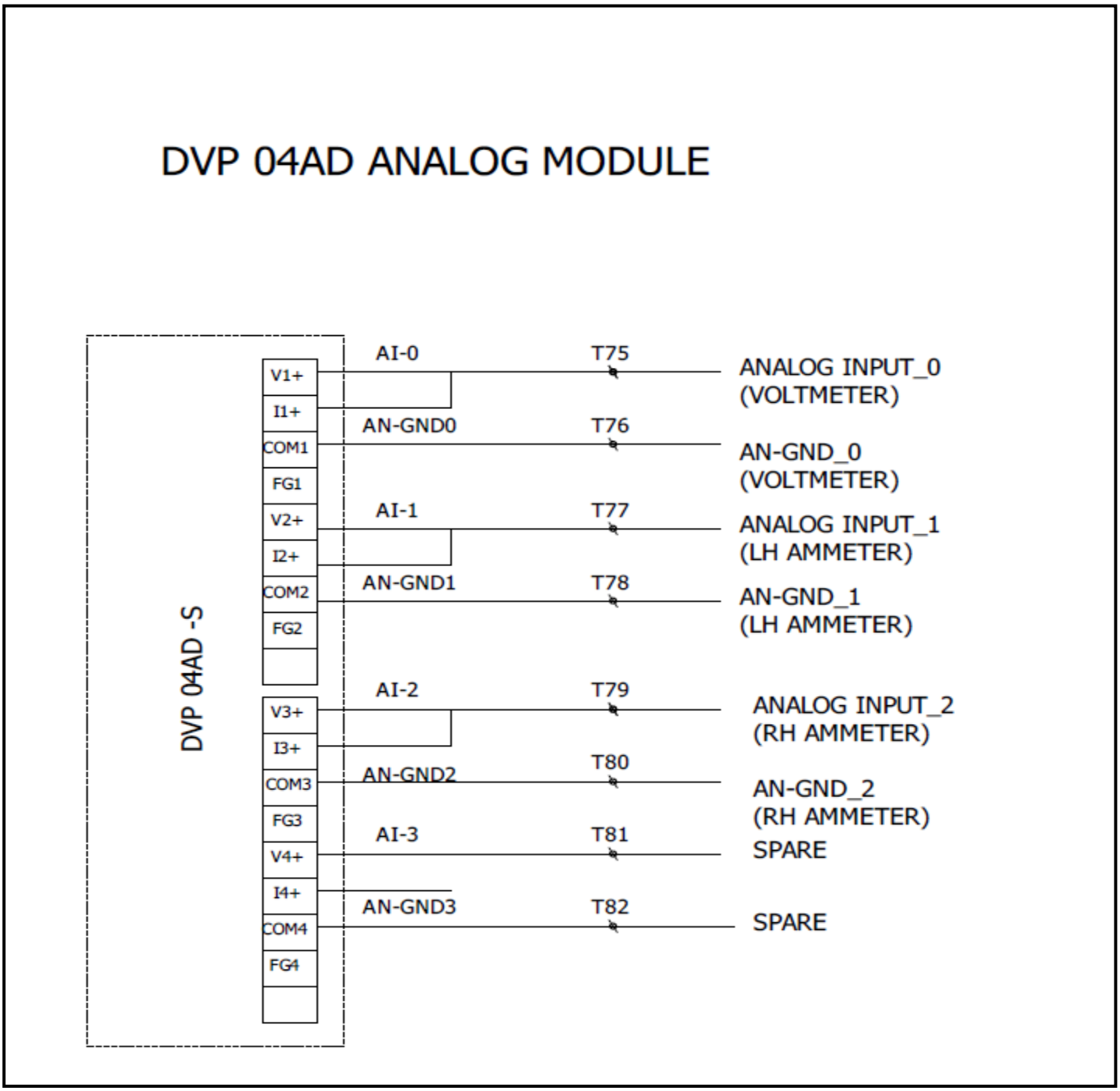
EXPANSION MODULE 2: DVP 16SP11R-2 CONTROL DIAGRAM



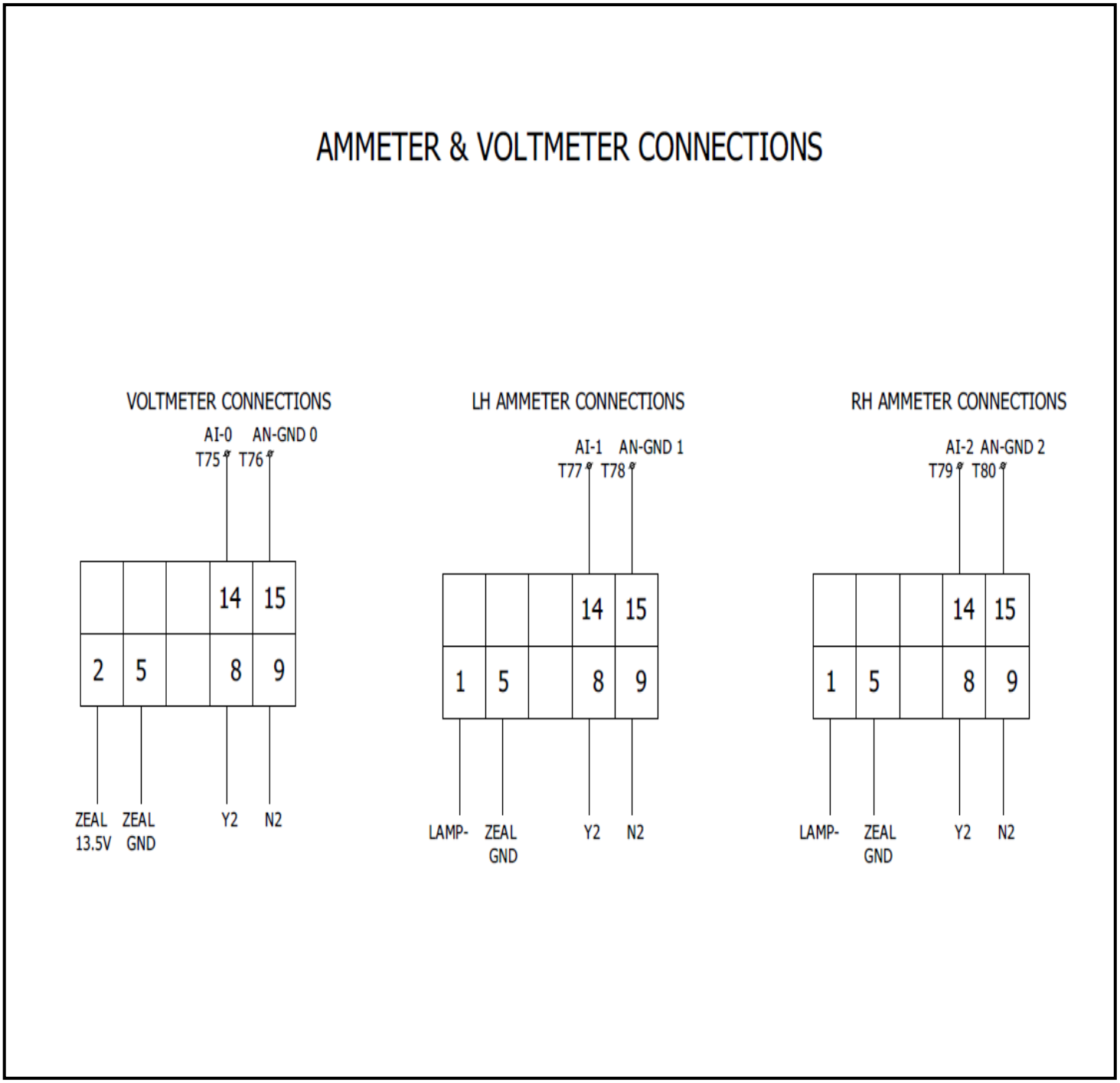
EXPANSION MODULE 3: DVP 08SN11R CONTROL DIAGRAM



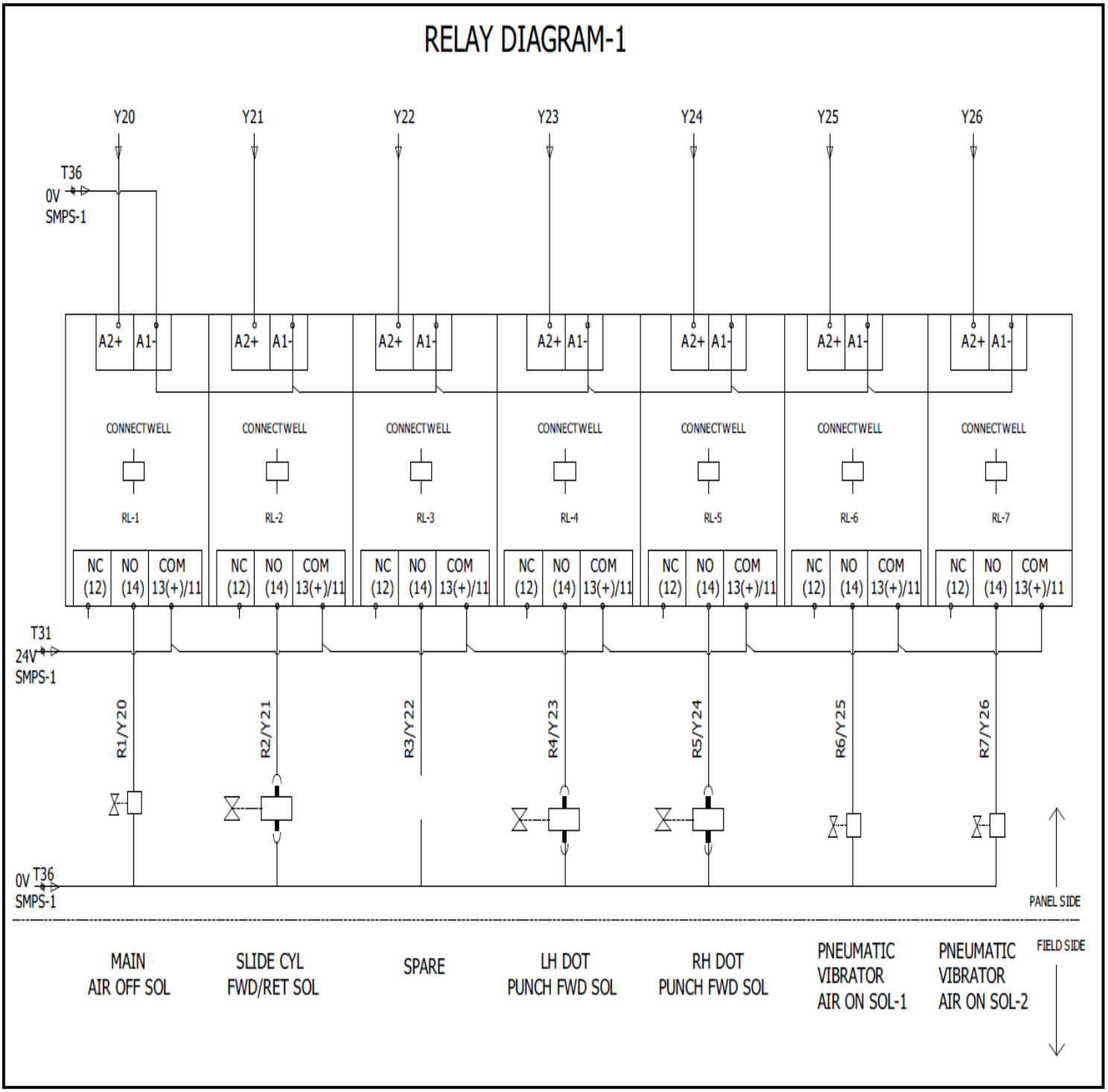
ANALOG MODULE: DVP 04AD-S



AMMETER & VOLTMETER CONNECTINS



RELAY DIAGRAM-1



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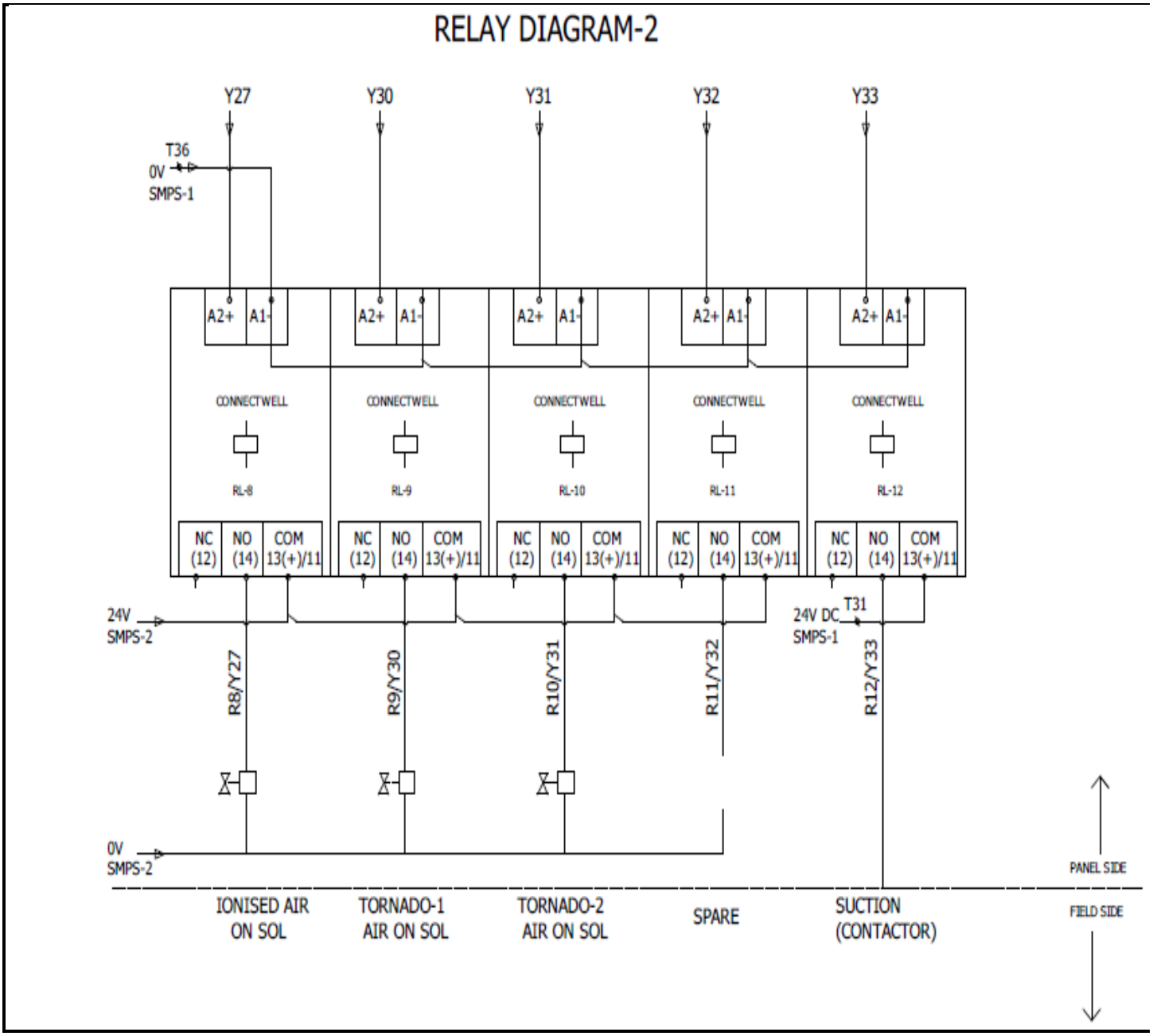
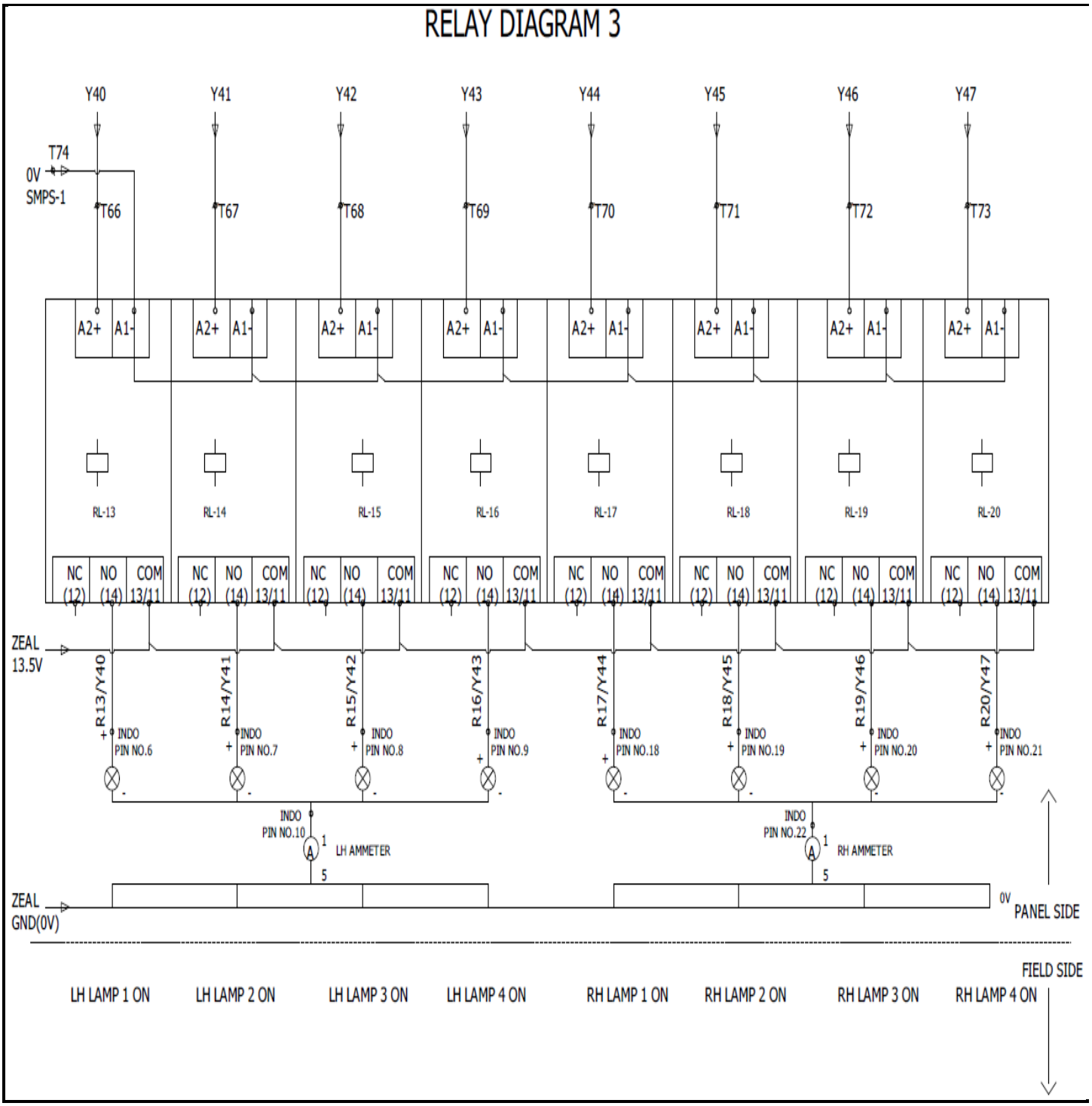
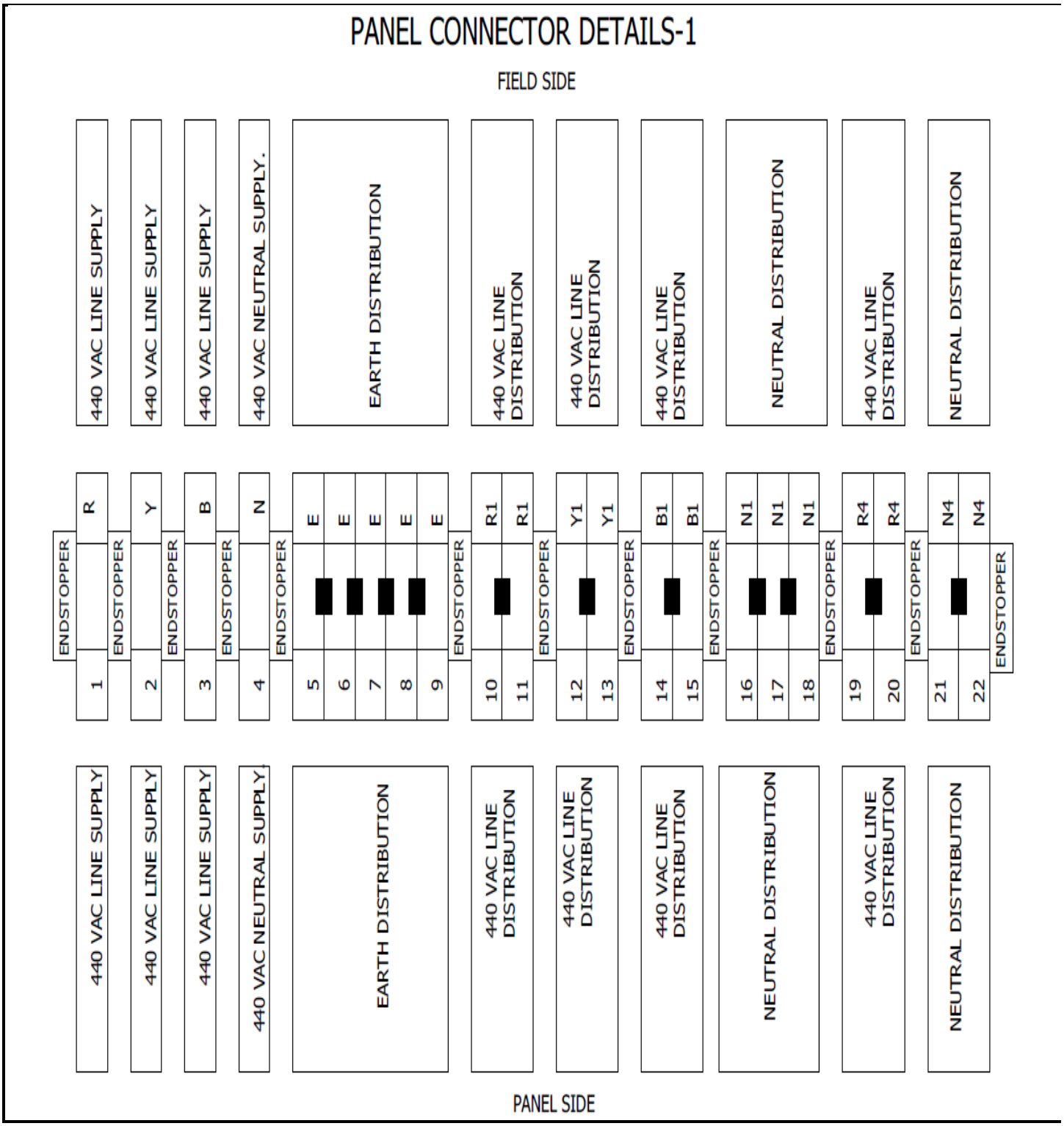


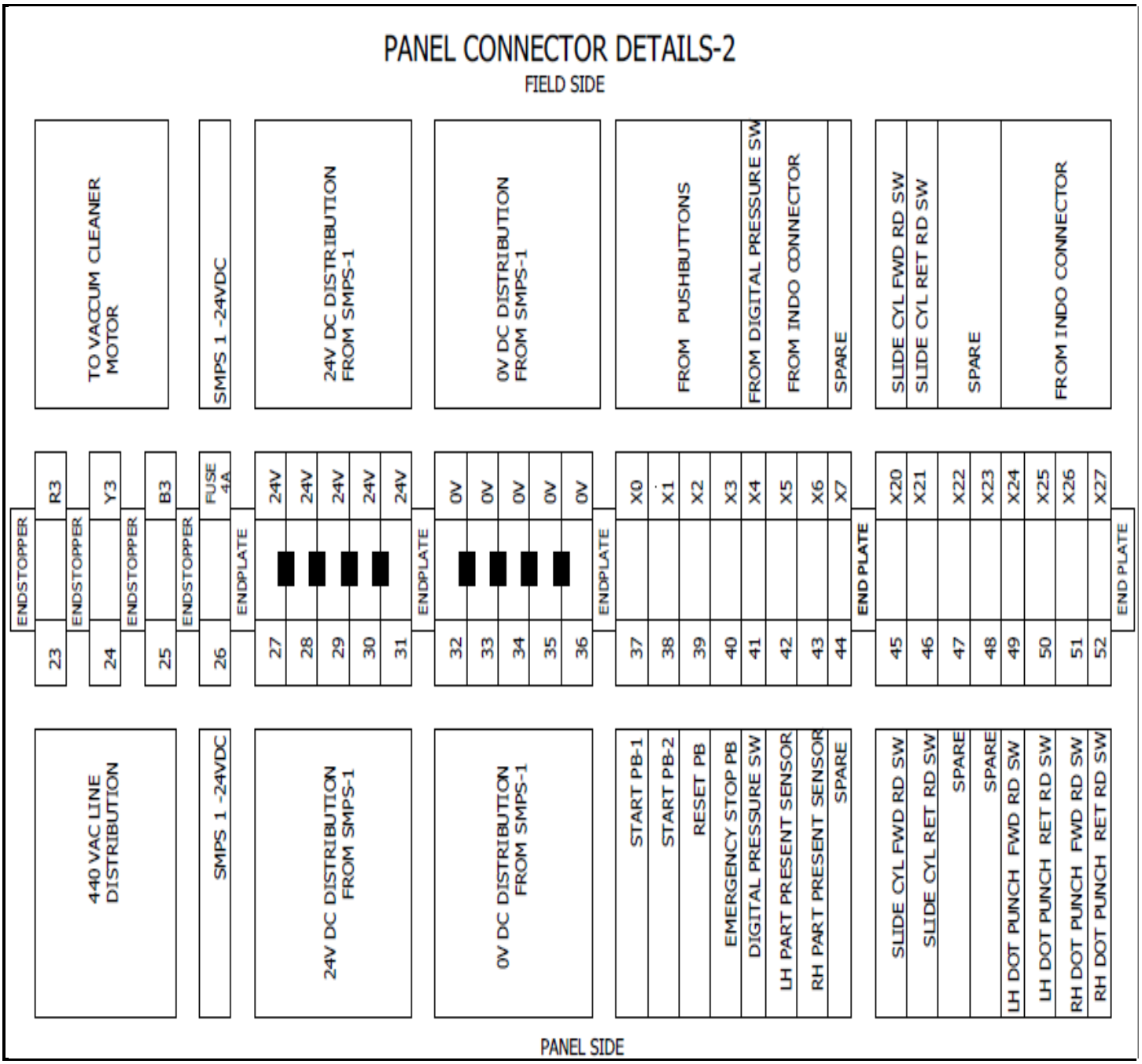
RELAY DIAGRAM-3



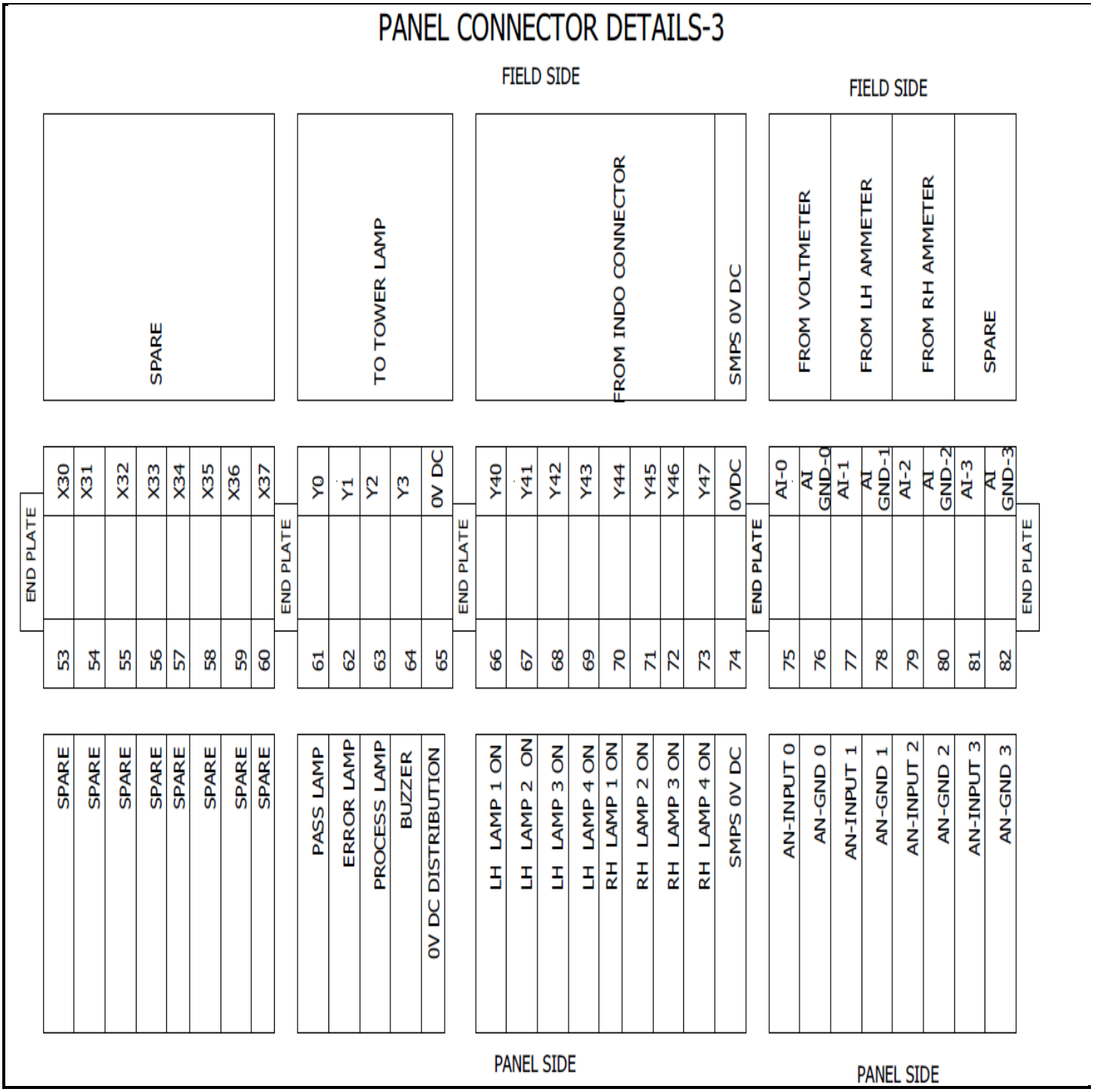
PANEL CONNECTOR DETAILS-1



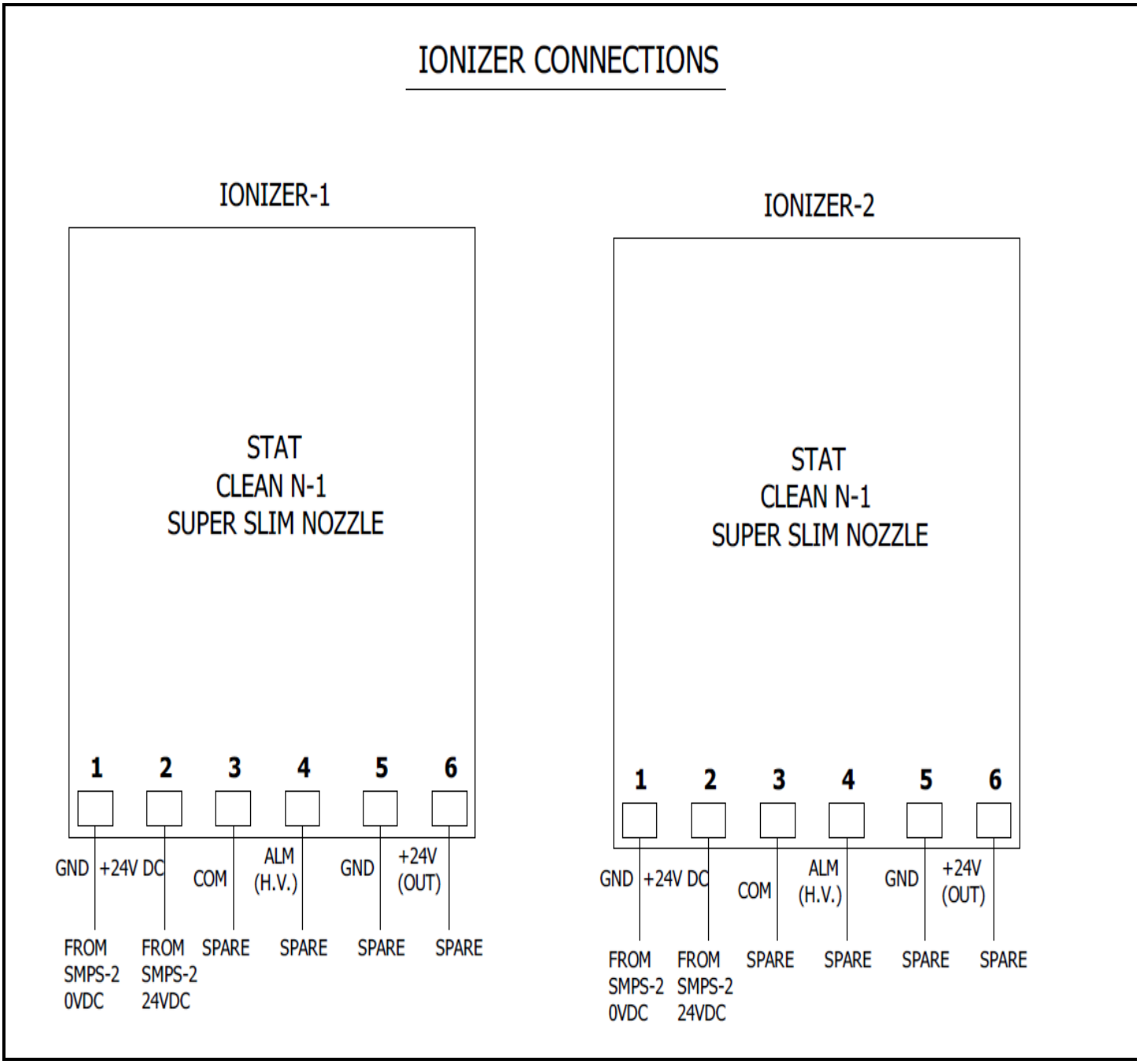
PANEL CONNECTOR DETAILS-2



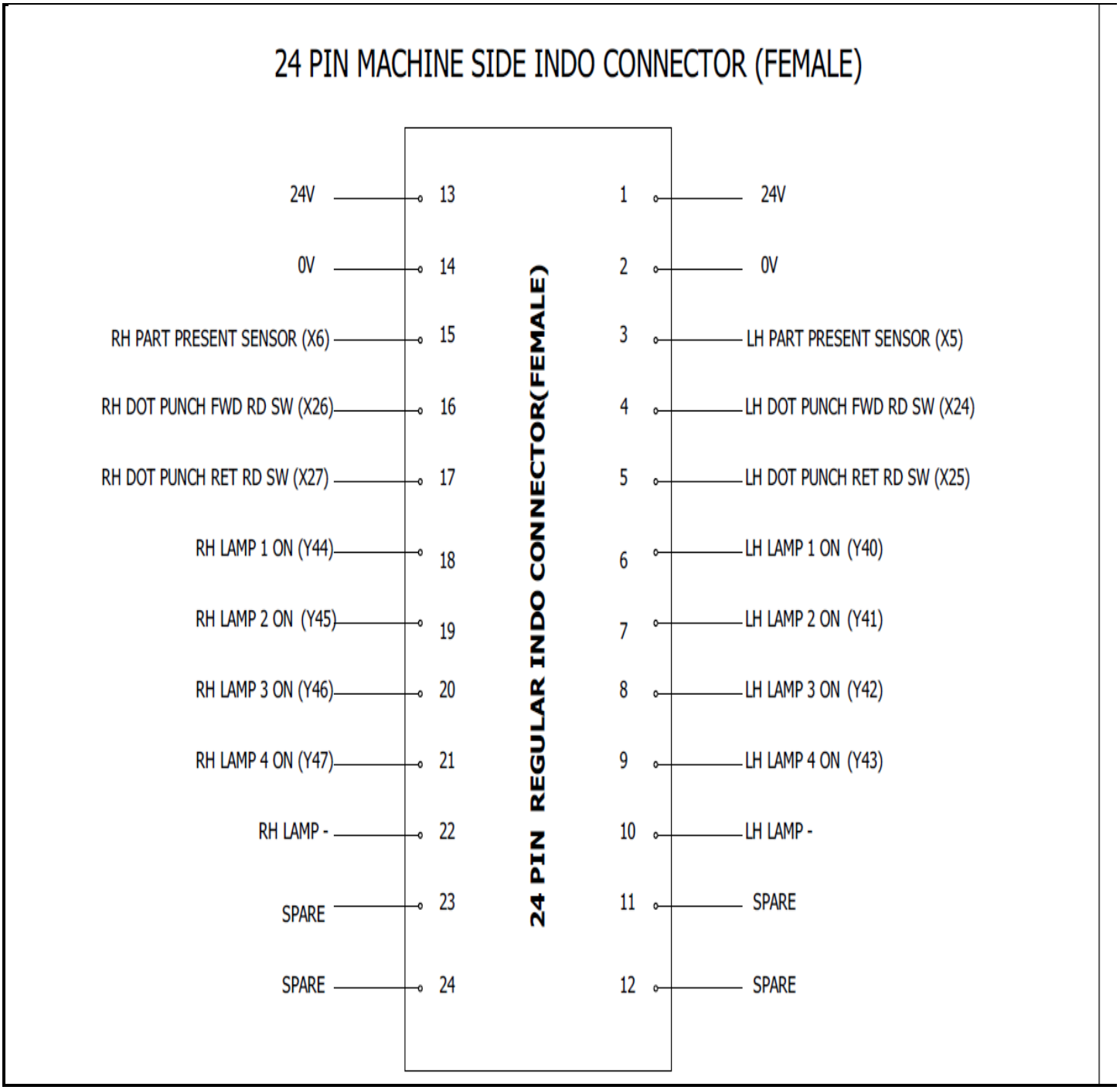
PANEL CONNECTOR DETAILS-3



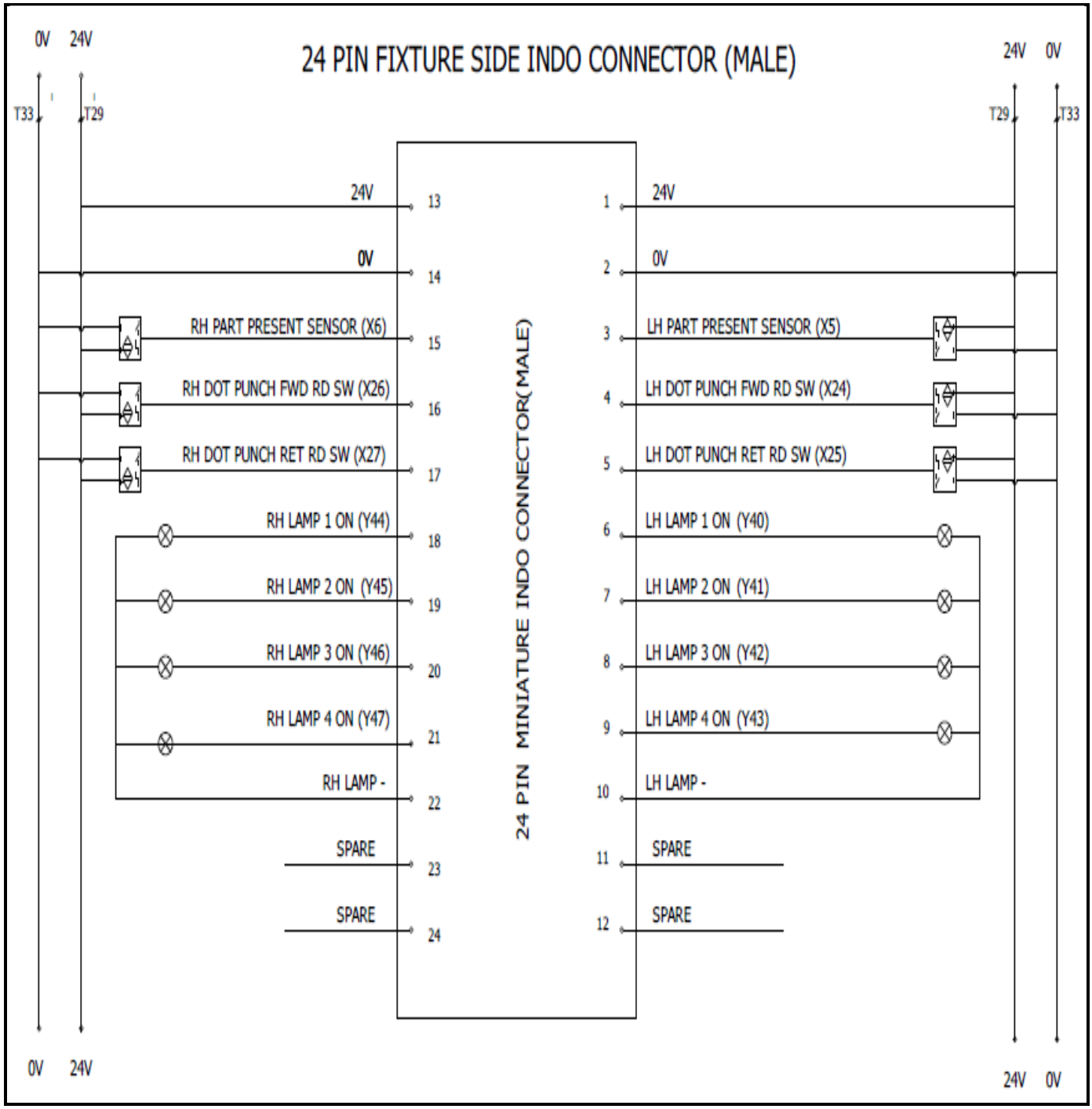
IONISED BLOWER CONNECTIONS



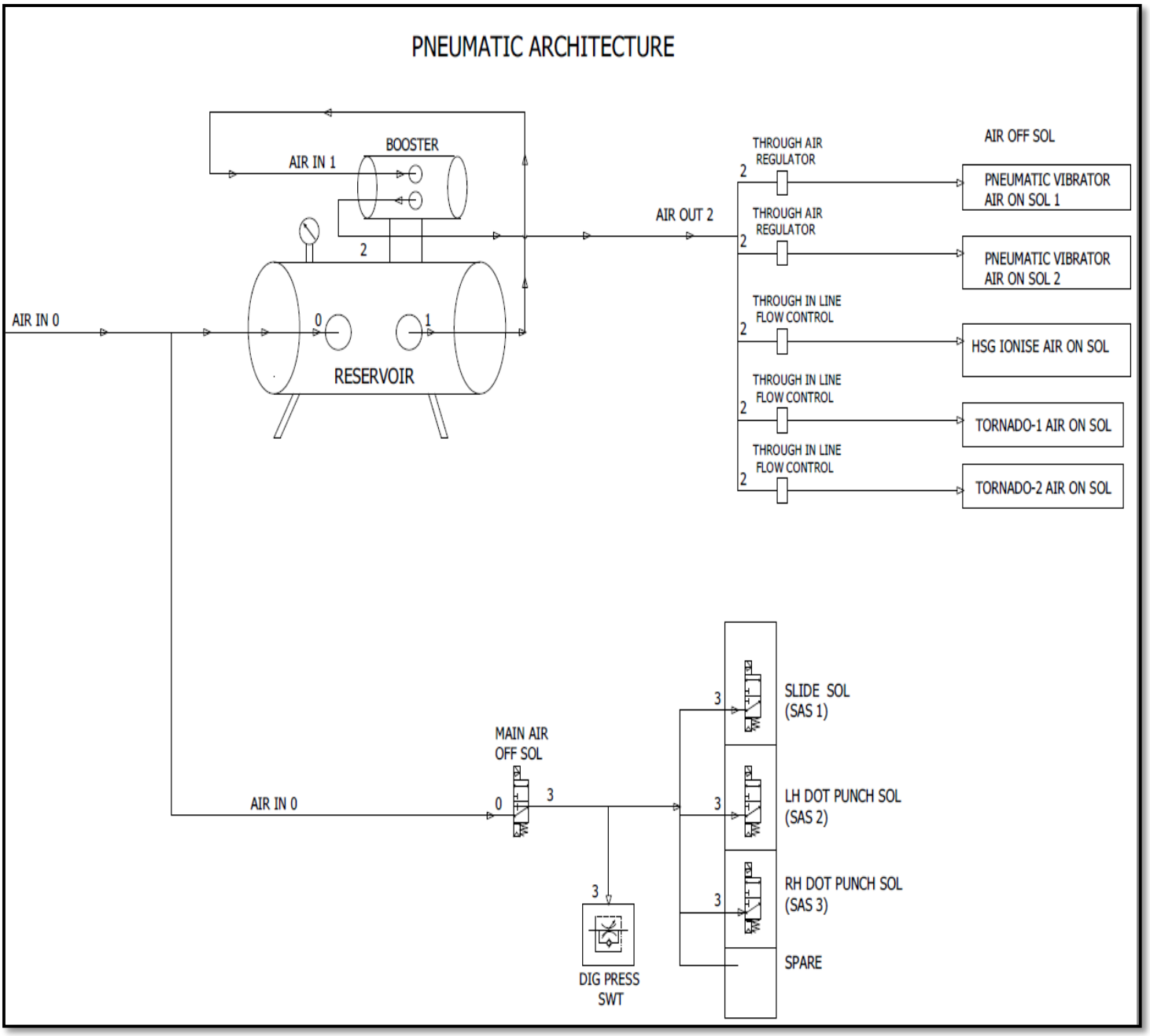
MACHINE SIDE 24 PIN INDO CONNECTOR (FEMALE)



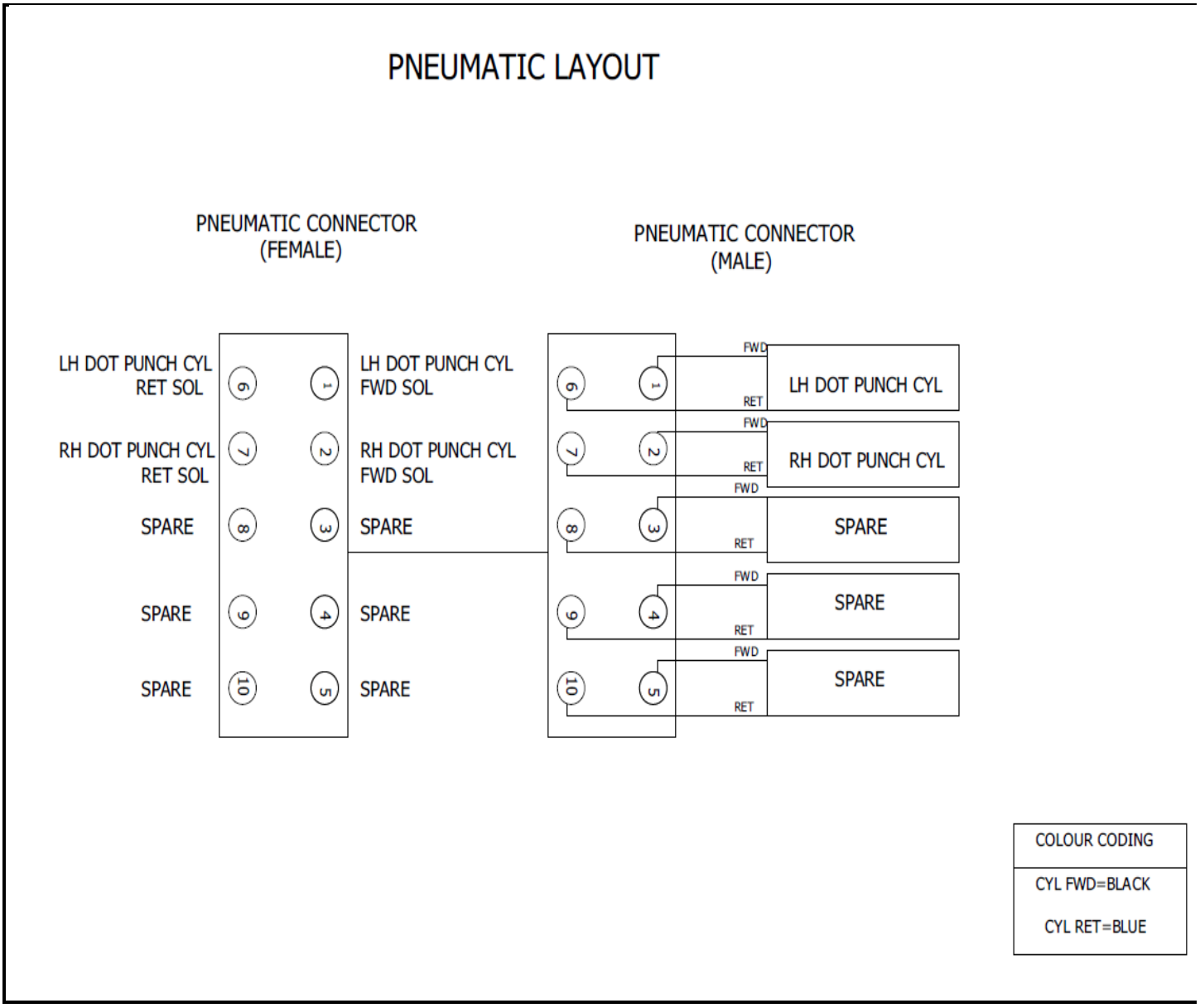
FIXTURE SIDE 24 PIN INDO CONNECTOR (MALE)



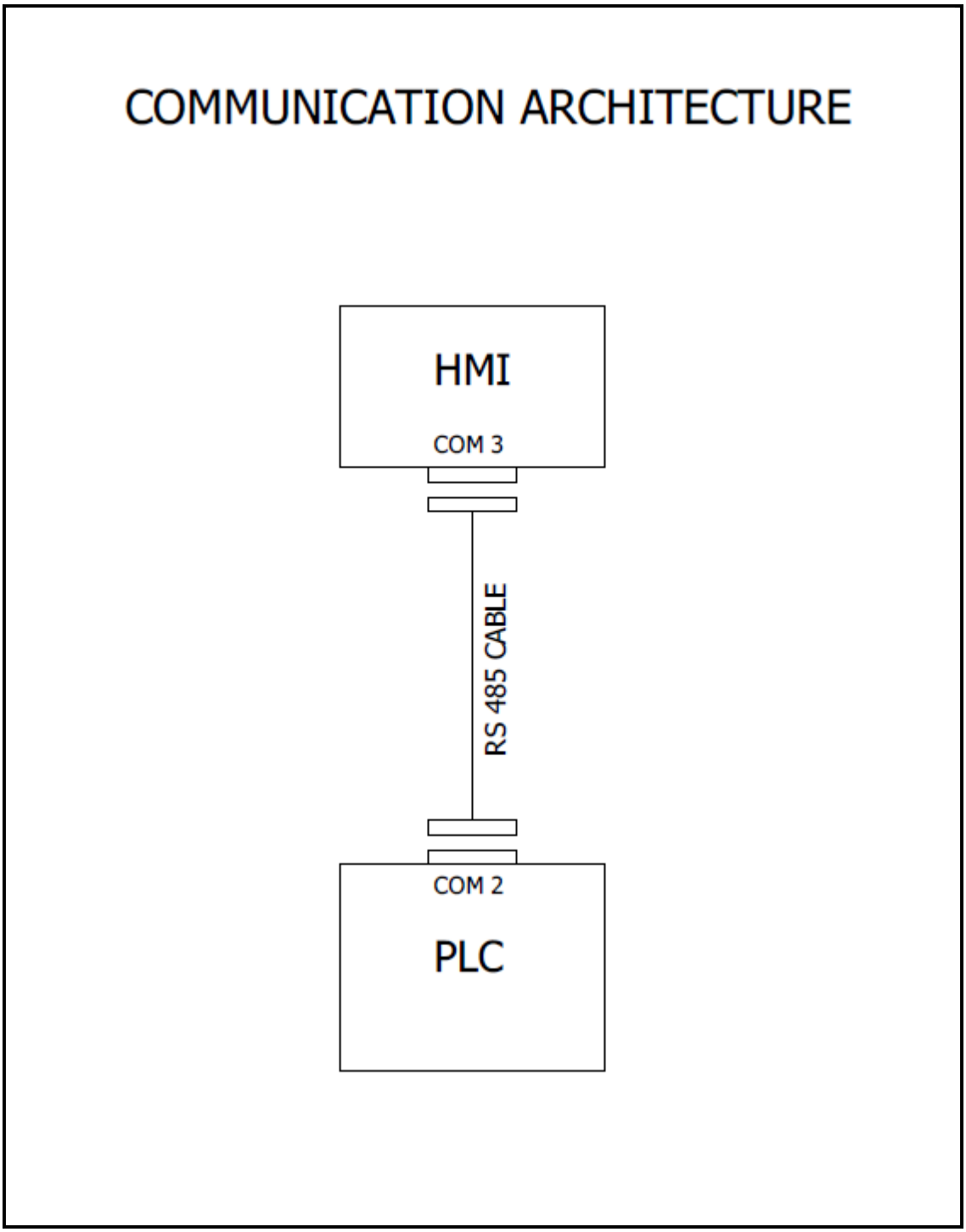
PNEUMATIC CONNECTIONS



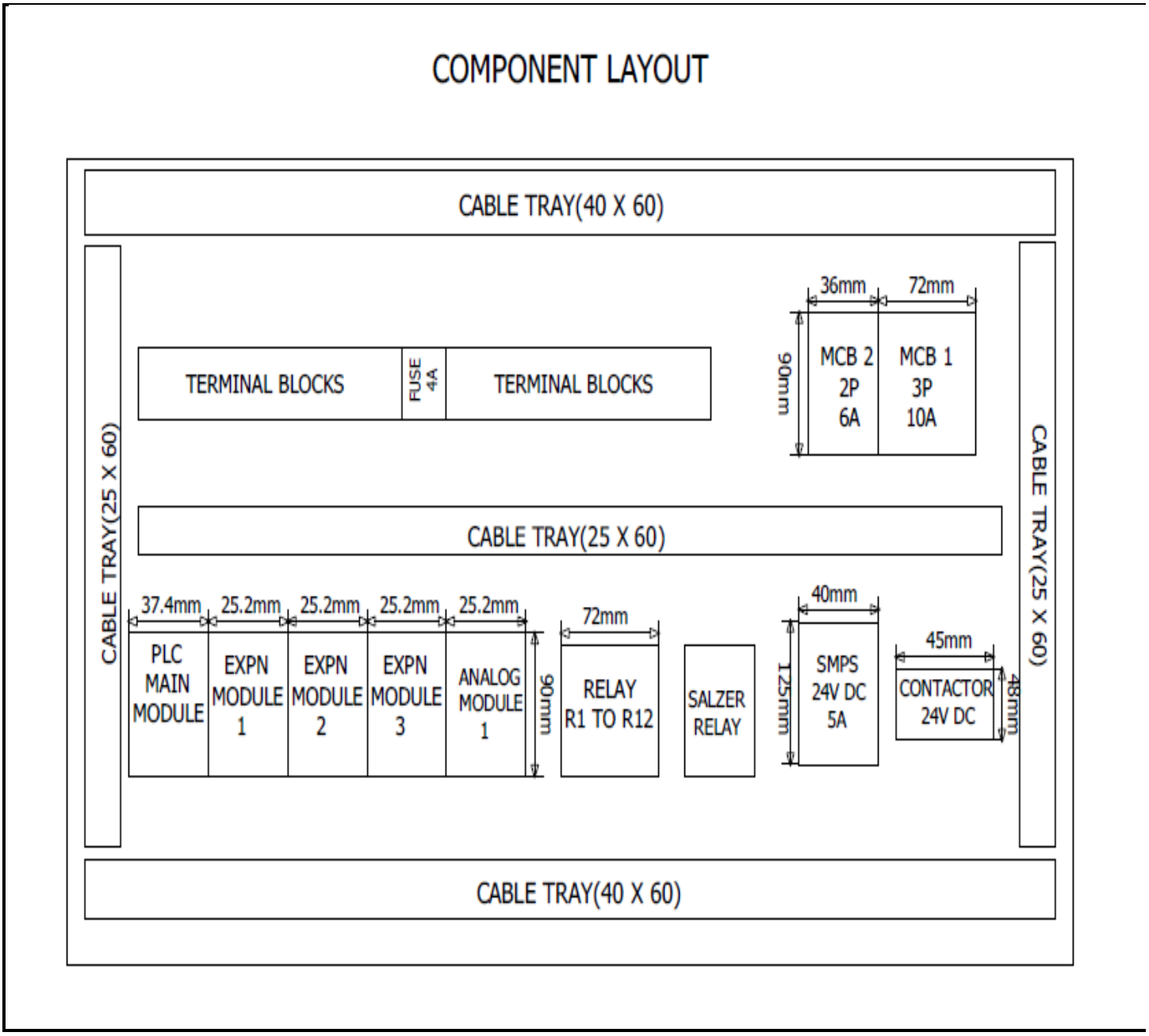
PNEUMATIC LAYOUT



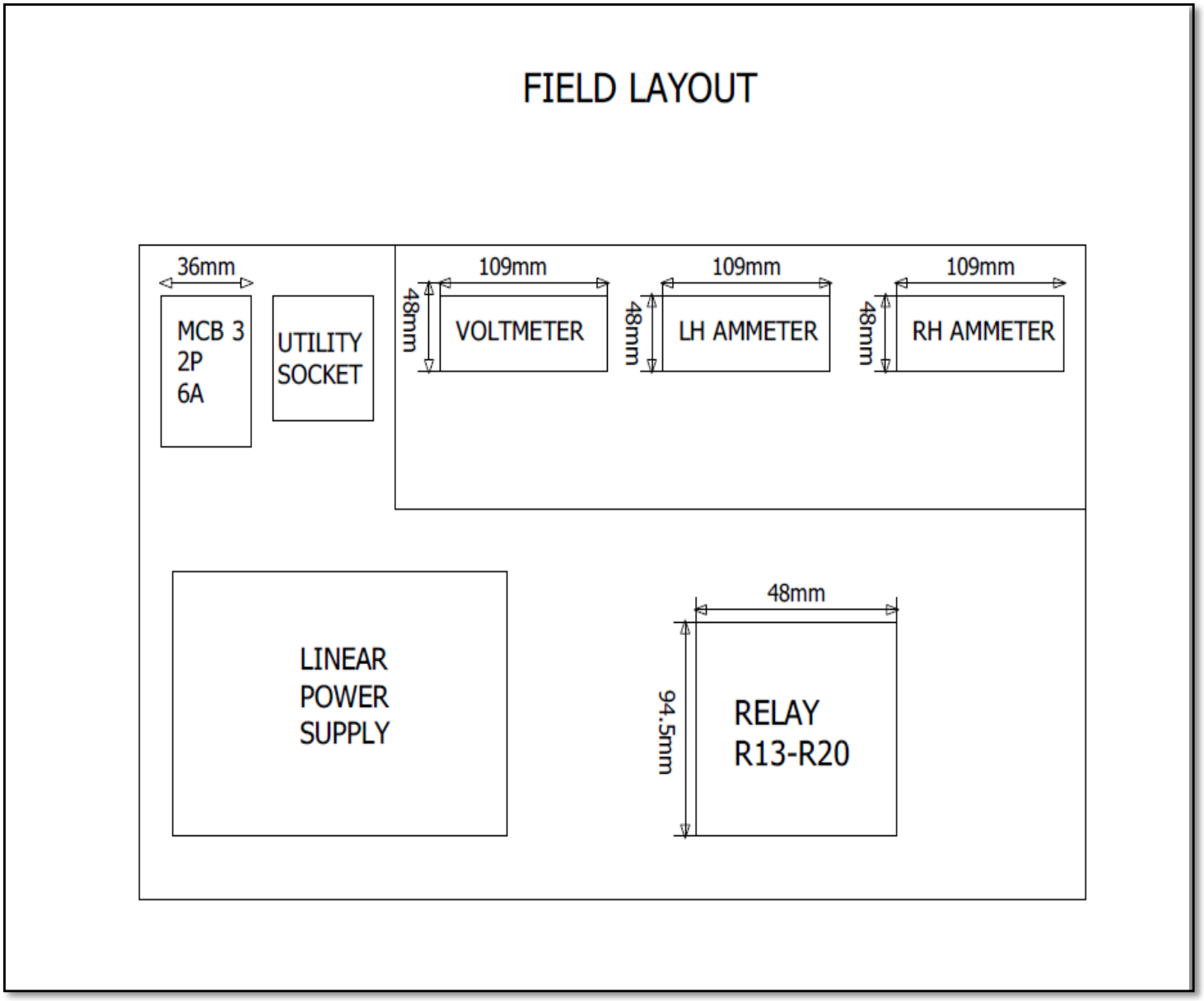
COMMUNICATION ARCHITECTURE



COMPONENT LAYOUT

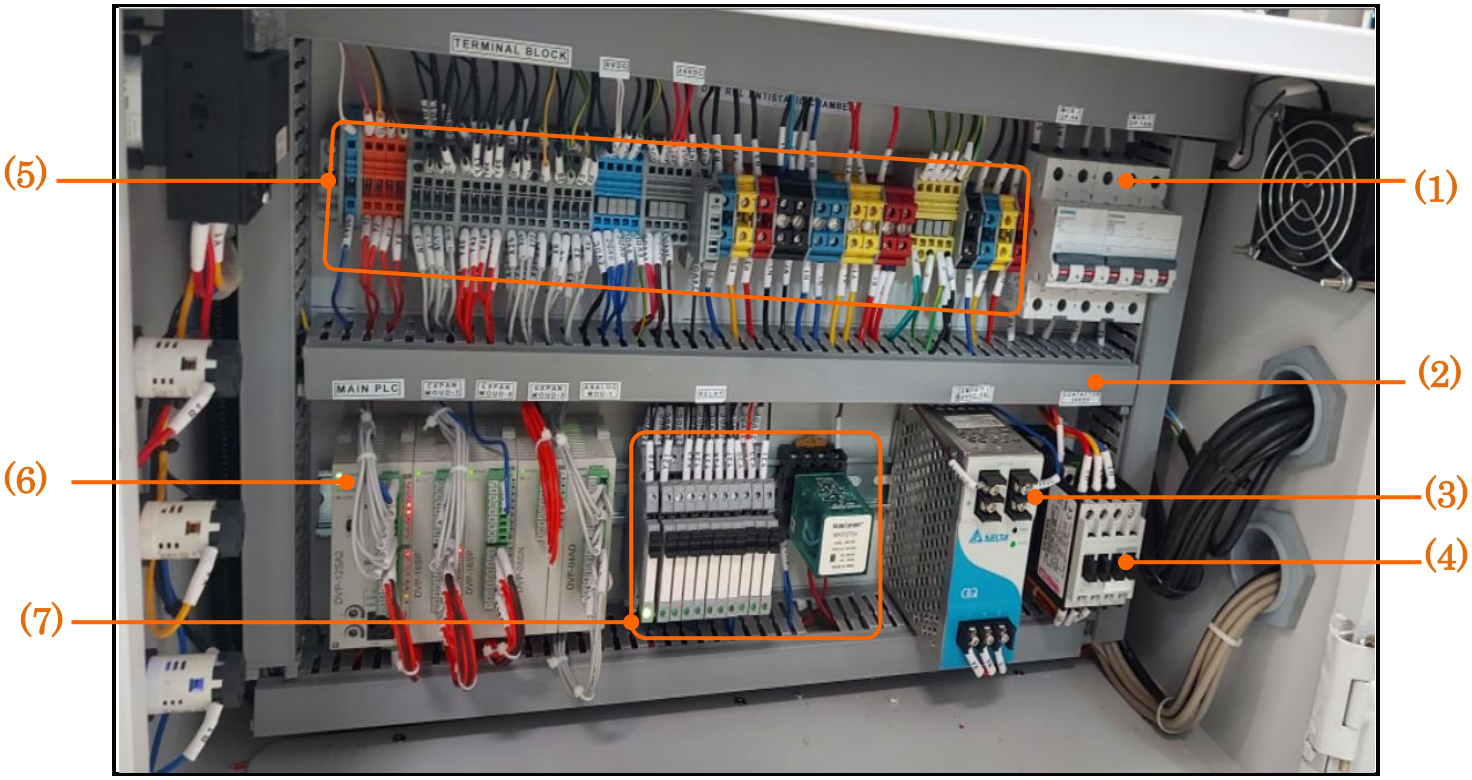


FIELD LAYOUT



Component Layout

Internal Panel Layout



No.	Component Name	Description
1	MCB	To Protection against short circuit and overloading current.
2	Cable Tray	To Route Internal Wires.
3	SMPS (24V, 5 A)	To Power control panel.
4	Contactor	A Contactor is an electrical device which is used for switching an electrical circuit on or off.
5	Terminal blocks	To connect external devices.

6	PLC	1) 8DI/4DO, 2) 8DI/8DO,3) 8DI/8DO,4) 8DO
7	1-channel Relays + SALZER Relay	To connect solenoid valves.

CABLE DETAILS

CABLE DETAILS

CABLE	CORE DIAMETER SHILDED/FLEXIBLE	COLOUR		CABLE
POWER CABLE	2.5 SQMM 4 CORE FLEXIBLE	RED	LINE	POLYCAB
		BLACK	NEUTRAL	
		YELLOW +GREEN	EARTHING	
PANNEL WIRING	CONTROL WIRING 0.5 SQMM 1-CORE	BLUE	(+) 24V DC	RRCABLE
		WHITE	(-) 0V DC	
		GRAY	I/P TO PLC	
		ORANGE	O/P TO PLC	
		PINK	(+) 12V DC	
		YELLOW	(-) 0V DC	
	POWER WIRING 2 SQMM 1-CORE	RED	LINE	RRCABLE
		BLACK	NEUTRAL	
		YELLOW +GREEN	EARTHING	
PLC TO HMI (RS 485) COMMUNICATION CABLE	2 CORE 0.25 SQMM SHILDED	BROWN	D+	STANDARD
		WHITE	D-	
TOWER LAMP	6 CORE, 0.25 SQMM SHILDED			STANDARD

BILL OF MATERIAL:

YFG FCL ANTISTATIC CHAMBER					
ELECTRICAL PANEL MATERIAL					
SR.NO	COMPONENT NAME	DESCRIPTION	PART NO.	MAKE	QTY.
1	Control panel	700X400X210	MFG700X400X210	MFG	1
2	Plc	8DI/4DO	DVP12SA211R	Delta	1
3	Expansion module	8DI/8DO	DVP16SP11R	Delta	2
4	Expansion module	8D0	DVP08SN11R	Delta	1
5	Expansion module	4 Analog input	DVP04AD	Delta	1
6	Hmi	7" Color	DOP107CV	Delta	1
7	Smps	24VDC,5A	DRL-24V120W1AA	Delta	1
8	Linear Power supply	0-15V,5A	202108181	Zeal	1
9	Relay	1-Channel,24VDC	SOCY1-24D	Connectwell	20
10	Mcb	2P,6A	5SL62067RC	SIEMENS	2
11	Mcb	3P,10A	5SL63107RC	SIEMENS	1
12	Power lamp (red)	230VAC	3PLBR4L	Teknic	1
13	Power lamp (yellow)	230VAC	3PLBR8L	Teknic	1
14	Power lamp (blue)	230VAC	3PLBR6L	Teknic	1
15	Push button (green)	Mushroom Type	2AM3	Teknic	2
16	Emergency stop (red)	Latch Type	2AML4	Teknic	1
17	No elements	Green	S1	Teknic	2
18	Nc elements	Red	S2	Teknic	1
19	Pb box	2 Way Metal	50mm	Teknic	2
20	Contactor	24 VDC Coil	3TF30 10 OBB4	Siemens	1
21	On/off switch	3P,32A	LB116	Salzer	1
22	Relay	10A,24V DC	MKP2P24	Salzer	1
23	Voltmeter	230VAC	MT4W-DV	Autonics	1
24	Ammeter	230VAC	MT4W-DA	Autonics	2
26	Electrical connector	16PIN (MALE)	0110016M011	Indo Electric	1
		16 PIN (FEMALE)	0110016F011		1
27	Industrial socket	3 phase,5 pin	TYP 215	Cosmos	1

Pneumatic Material					
Sr. No	**Component Name**	**Description**	**Part No.**	**Make**	**Qty.**
1	AIROFF	VUVS-L20-M32C-AD-G18-F7-1C1	575269	FESTO	5
2	FR UNIT	1/4"	AW20-02BG1	SMC	1
3	SOLENOID VALVE	SINGLE ACTING	SY7120-5LZ-02	SMC	3
4	Pneumatic Quick Coupler	10 port	KDM10-06	SMC	1
5	Digital Pressure SW	Pressure Display	ISE30AF-01-P-LA2	SMC	1
6	Pneumatic Vibrator	-	T16	VIBSOL	2
7	Pneumatic booster regulator	1/4"	VBA10A-02GN	SMC	1
8	Pneumatic reservoir	10 LITRE,AIR TANK	-	SMC	1
9	Manifold	4 way	SMC SS5Y7-20-04	SMC	1
10	Reed switch	3 wire,24v dc	D-M9PL-LUG	SMC	6
11	Pressure Regulator	1/4"	AR20-02BG1-B	SMC	1
12	MAIN AIR OFF	3 PORT	VT307-5G1-02F	SMC	1

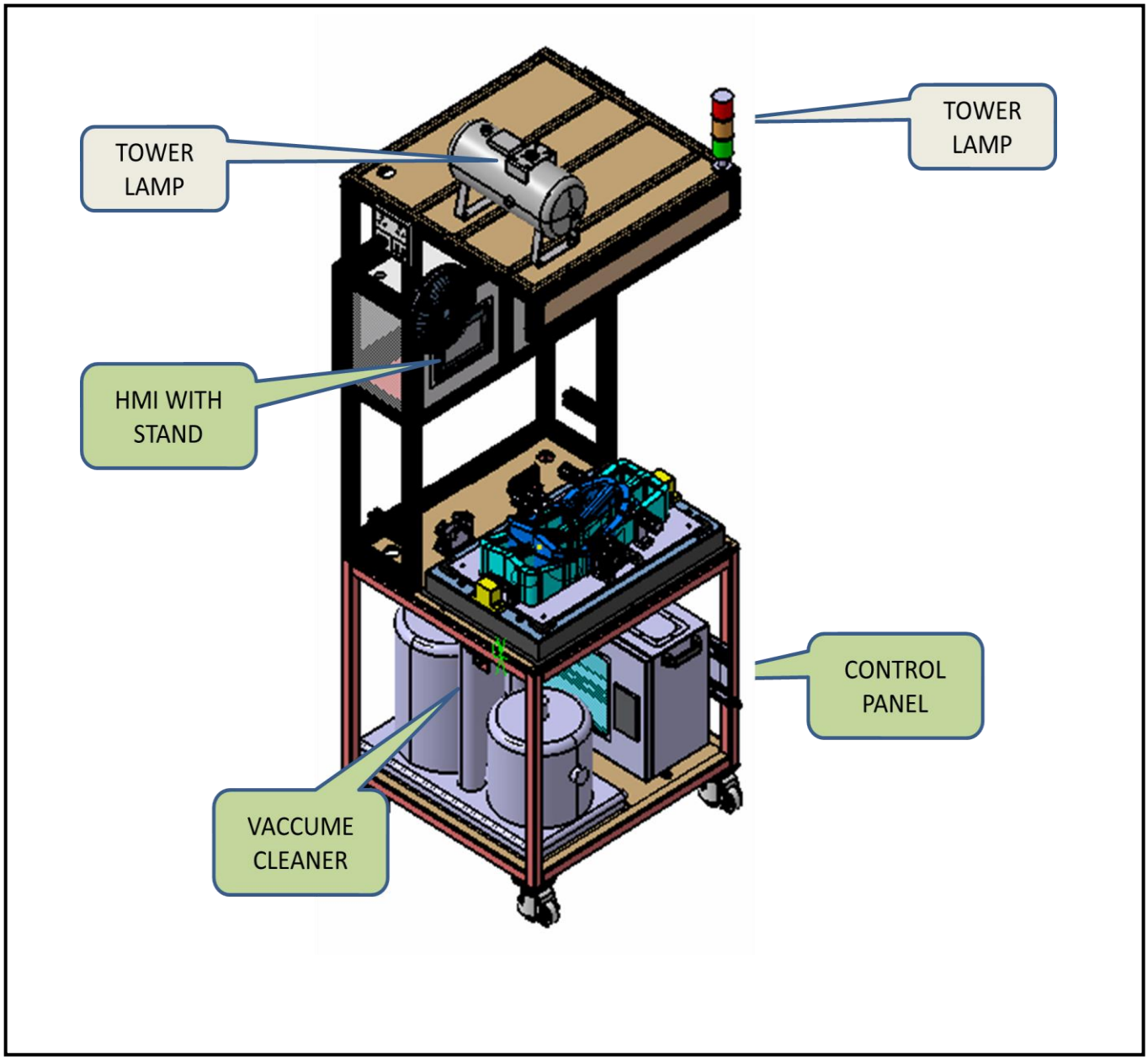
I/O LIST:

MODULE	ADDR	DESCRIPTION	ADDR	DESCRIPTION	RELAY
DVP12SA211R	X0	START PB 1	Y0	PASS LAMP	
	X1	START PB 2	Y1	ERROR LAMP	
	X2	RESET PB	Y2	PROCESS LAMP	
	X3	EMG STOP PB	Y3	BUZZER	
	X4	Digital PRESSURE S/W	Y4	NA	
	X5	LH Part Present Sensor	Y5	NA	
	X6	RH Part Present Sensor	Y6	NA	
	X7	SPARE	Y7	NA	
DVP16SP11R-1	X20	SLIDE CYL FWD RD SW	Y20	MAIN AIR OFF	R1
	X21	SLIDE CYL RET RD SW	Y21	SLIDE CYL FWD SOL	R2
	X22	SPARE	Y22	SPARE	R3
	X23	SPARE	Y23	LH DOT PUNCH CYL FWD SOL	R4
	X24	LH DOT PUNCH FWD RD SW	Y24	RH DOT PUNCH CYL FWD SOL	R5
	X25	LH DOT PUNCH RET RD SW	Y25	PNUMATIC VIBRATOR AIR OFF-1 SOL	R6
	X26	RH DOT PUNCH FWD RD SW	Y26	PNUMATIC VIBRATOR AIR OFF-2 SOL	R7

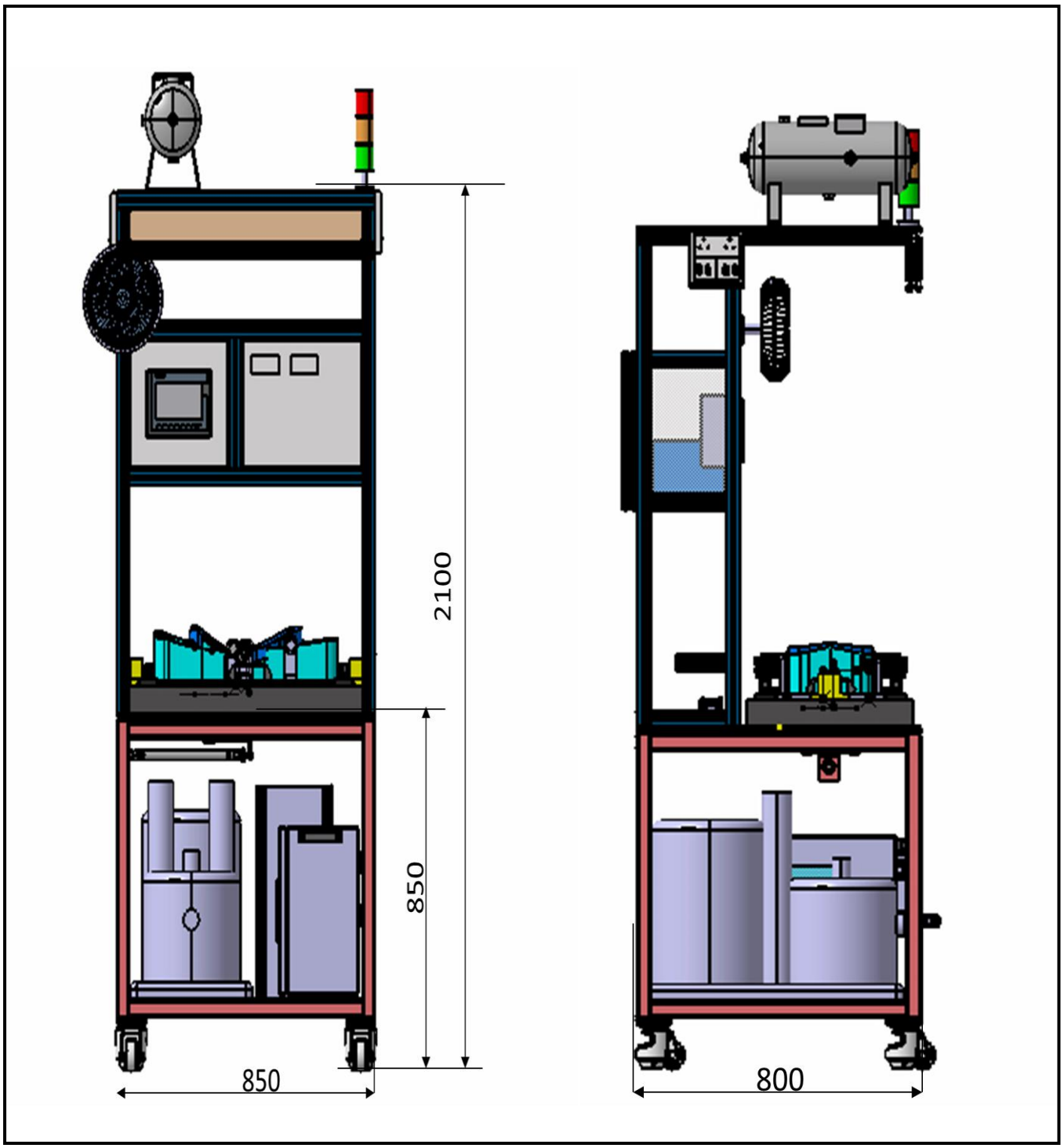
	X27	RH DOT PUNCH RET RD SW	Y27	SPARE	R8
DVP16SP11R-2	X30	SPARE	Y30	Tornado-1 AIR ON SOL	R9
	X31	SPARE	Y31	Tornado-2 AIR ON SOL	R10
	X32	SPARE	Y32	SPARE	R11
	X33	SPARE	Y33	SUCTION ON/OFF CONTACTOR	R12
	X34	SPARE	Y34	SPARE	
	X35	SPARE	Y35	Ionised Blower ON	Salzer relay
	X36	SPARE	Y36	SPARE	
	X37	SPARE	Y37	SPARE	
DVP08SN11R-3		NA	Y40	LH LAMP 1 ON	R13
		NA	Y41	LH LAMP 2 ON	R14
		NA	Y42	LH LAMP 3 ON	R15
		NA	Y43	LH LAMP 4 ON	R16
		NA	Y44	RH LAMP 1 ON	R17
		NA	Y45	RH LAMP 2 ON	R18
		NA	Y46	RH LAMP 3 ON	R19
		NA	Y47	RH LAMP 4 ON	R20
DVP 04AD-S	AI0	VOLTMETER			
	AI1	LH AMMETER			
	AI2	RH AMMETER			
	AI3	SPARE			

WORKSTATION DETAILS:

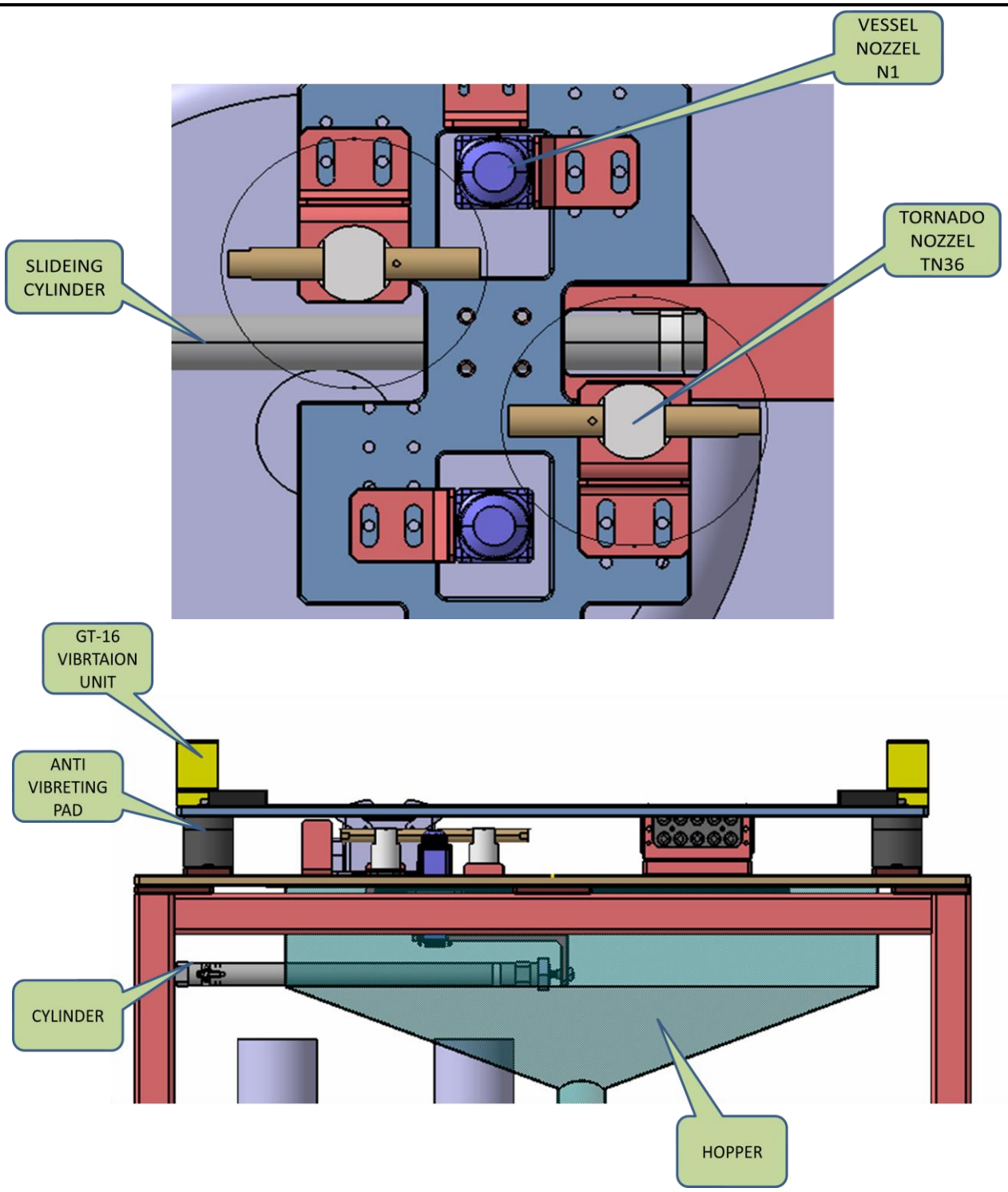
COMPONENT DESCRIPTIONTION



OVERALL DIMENSION

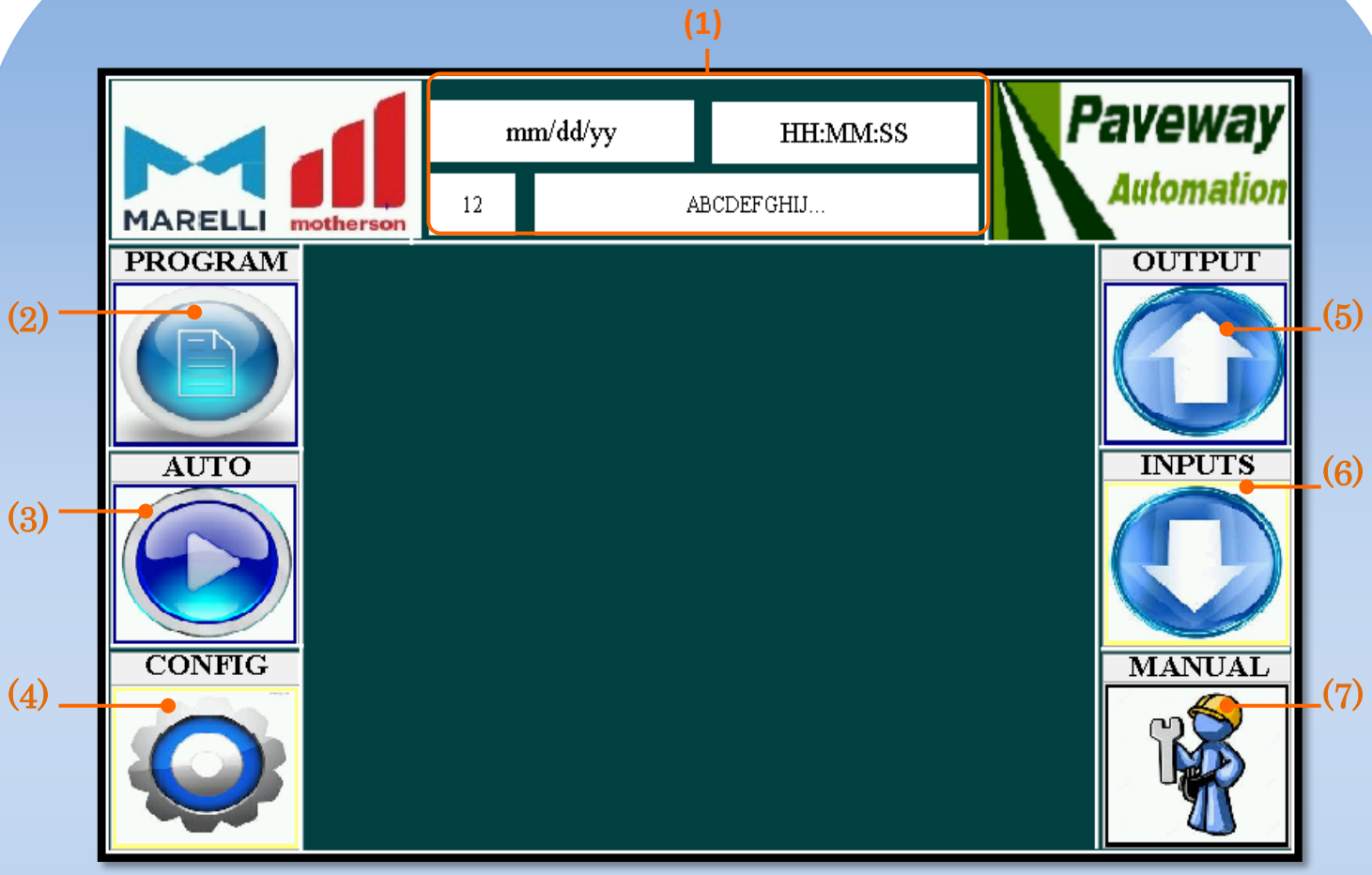


MACHINE PARTS DETAILS



HMI SCREEN DETAILS:

HOME SCREEN



1. Information Display Area
   Displays the Program name, Date & Time of currently in operation.

2.Program selection screen button .

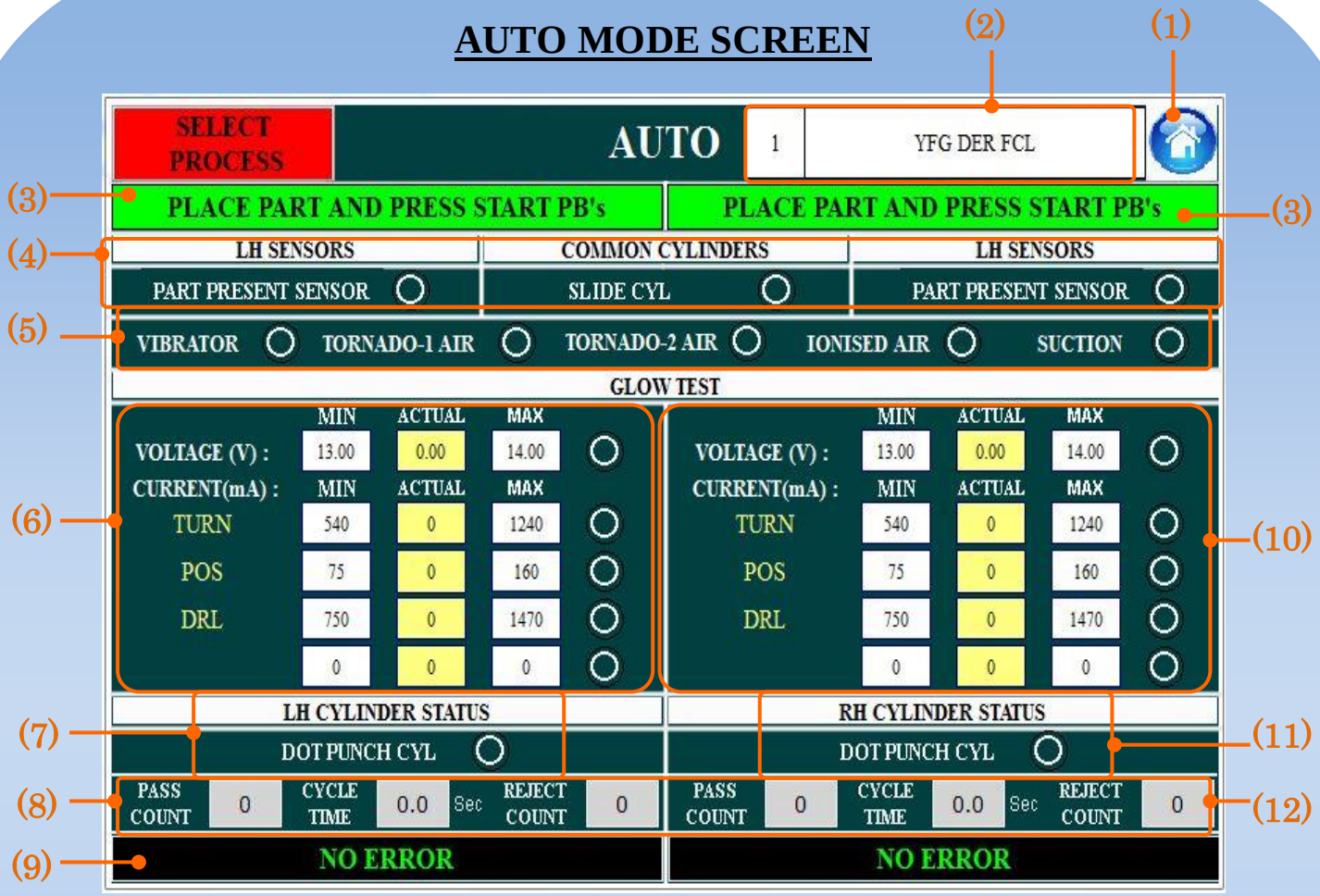
3.Auto monitering screen button.

4.Configuration screen button.

5.Output monitering screen button.

6.Input monitering screen button. 7. Manual operation screen button.

AUTO MODE SCREEN



1. Home screen navigation button.
2. Program name & number display.
3. Status message display.
4. Part present sensor & Cylinder operation status display.
5. Vibrator , Ionised air, Tornado air, External air, Suction operation status display.
6. GLOW TEST:-
   I) LH Side Actual,minimum & maximum voltage value display.

II) LH Side Lamp-1,Lamp-2,Lamp-3,Lamp-4 Current actual,minimum & maximum value display.

7. LH Side Dot punch cylinder operation status display.

8. LH Side Displays no. of parts passed and Reject count until last Reset & cycle time.

9. Error message display.

10. GLOW TEST:-

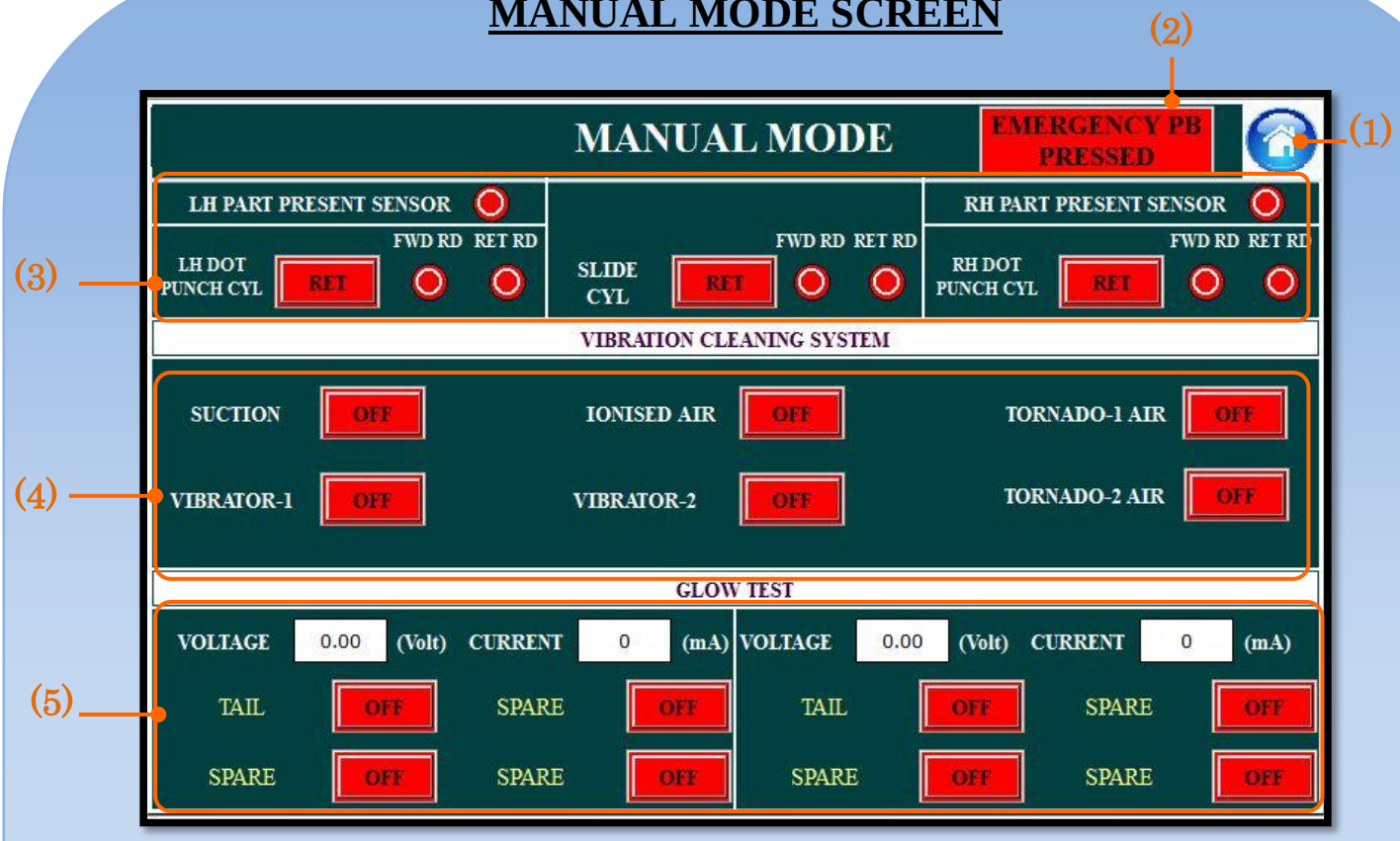
I) RH Side Actual,minimum & maximum voltage value display.

II) RH Side Lamp-1,Lamp-2,Lamp-3,Lamp-4 Current actual,minimum & maximum value display.

11. RH Side Dot punch cylinder operation status display.

12. RH Side Displays no. of parts passed and Reject count until last Reset & cycle time.

MANUAL MODE SCREEN



1. Home screen navigation button.

2. Emergency pushbutton status display (Pressed/Undressed)

3. i) LH & RH Part Present Sensor status display.

   ii) LH & RH Dot punch Cylinder manually pressed fwd./ret status display.

   iii) Slide cylinder manually pressed fwd./ret status display.

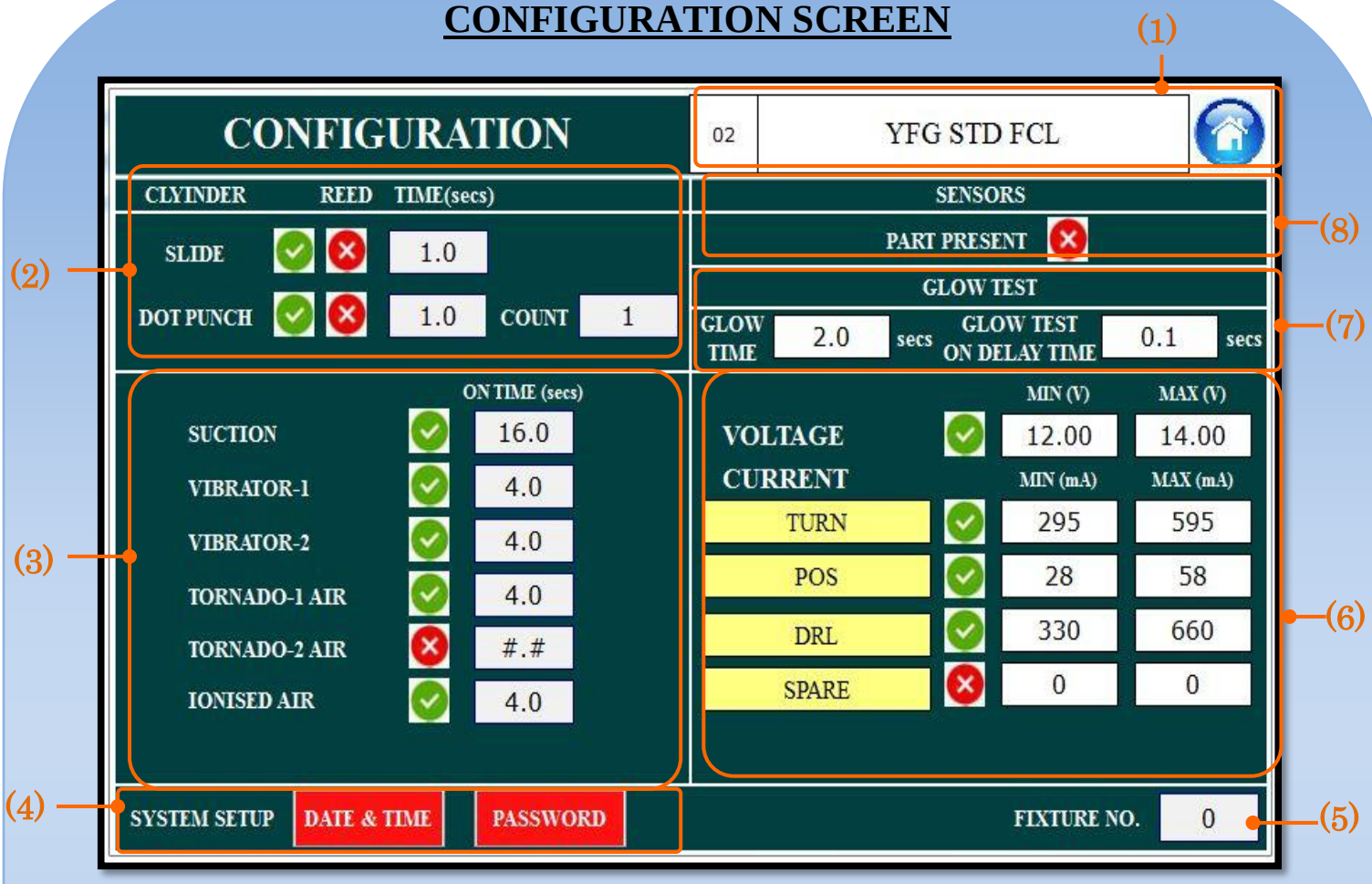
4. Vibration Cleaning System: -

   Manual ON/OFF Suction, Vibrator-1, Vibrator-2, Ionizing blower & Tornado-1 & 2 air.

5. Glow test LH & RH Side: -

   Manual ON/OFF LH & RH Side Lamp-1 ON, Lamp-2 ON, Lamp-3 ON, Lamp-4 ON buttons & its present voltage and current value display.

CONFIGURATION SCREEN



1. Home screen navigation button, program name & number display.

2. Cylinders Setting: -

   Slide cylinder & reed switch select/unselect option and setting time in sec.

   Dot punch cylinder & reed switch select/unselect option and setting time in sec.

   & punch count setting

3. Suction, Vibrator-1, Vibrator-2, Tornado -1 & 2 air, Ionized Blower Select/unselect option and setting time in sec.

4. System Password, Date & time setting buttons.

5. Fixture number selection option.

6. Lamp-1(TURN), Lamp-2(POS) & Lamp-3 (DRL) select/unselect option & its minimum - maximum voltage and current value setting option.

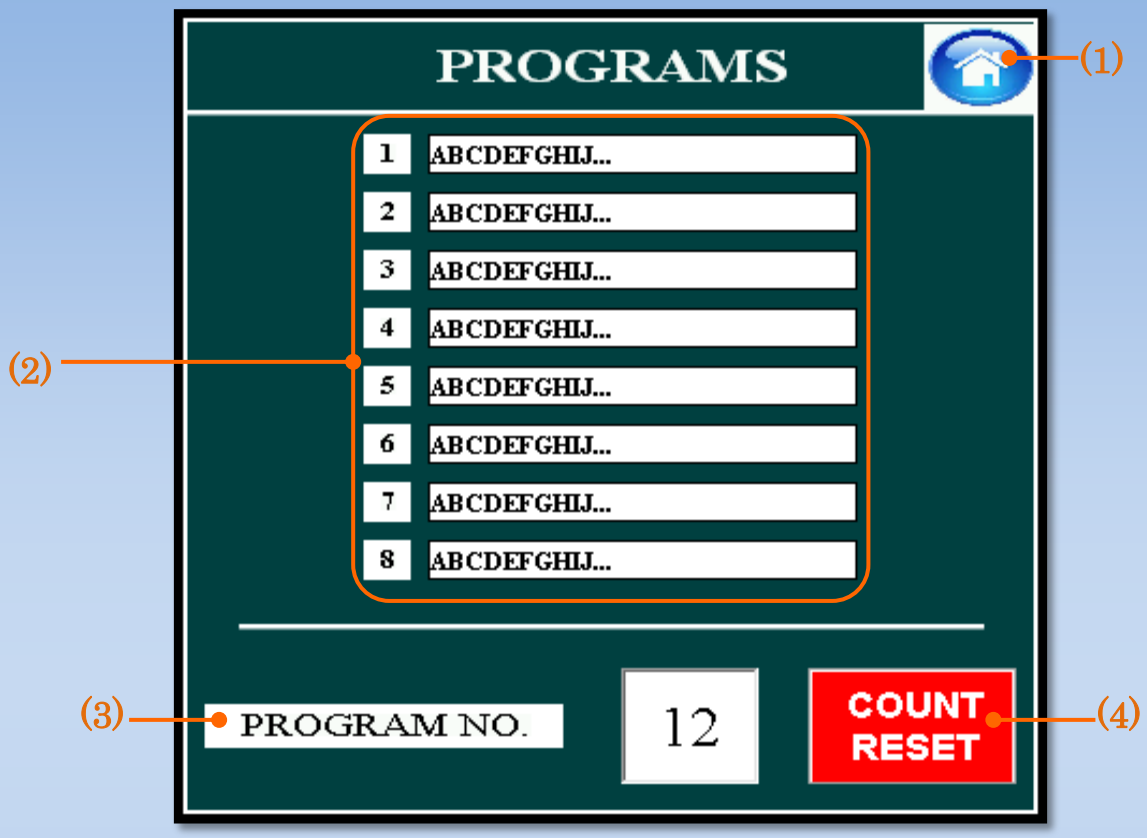
7. Glow test: -

   Glow test time in sec setting option.

   Glow test on delay time in sec setting option.

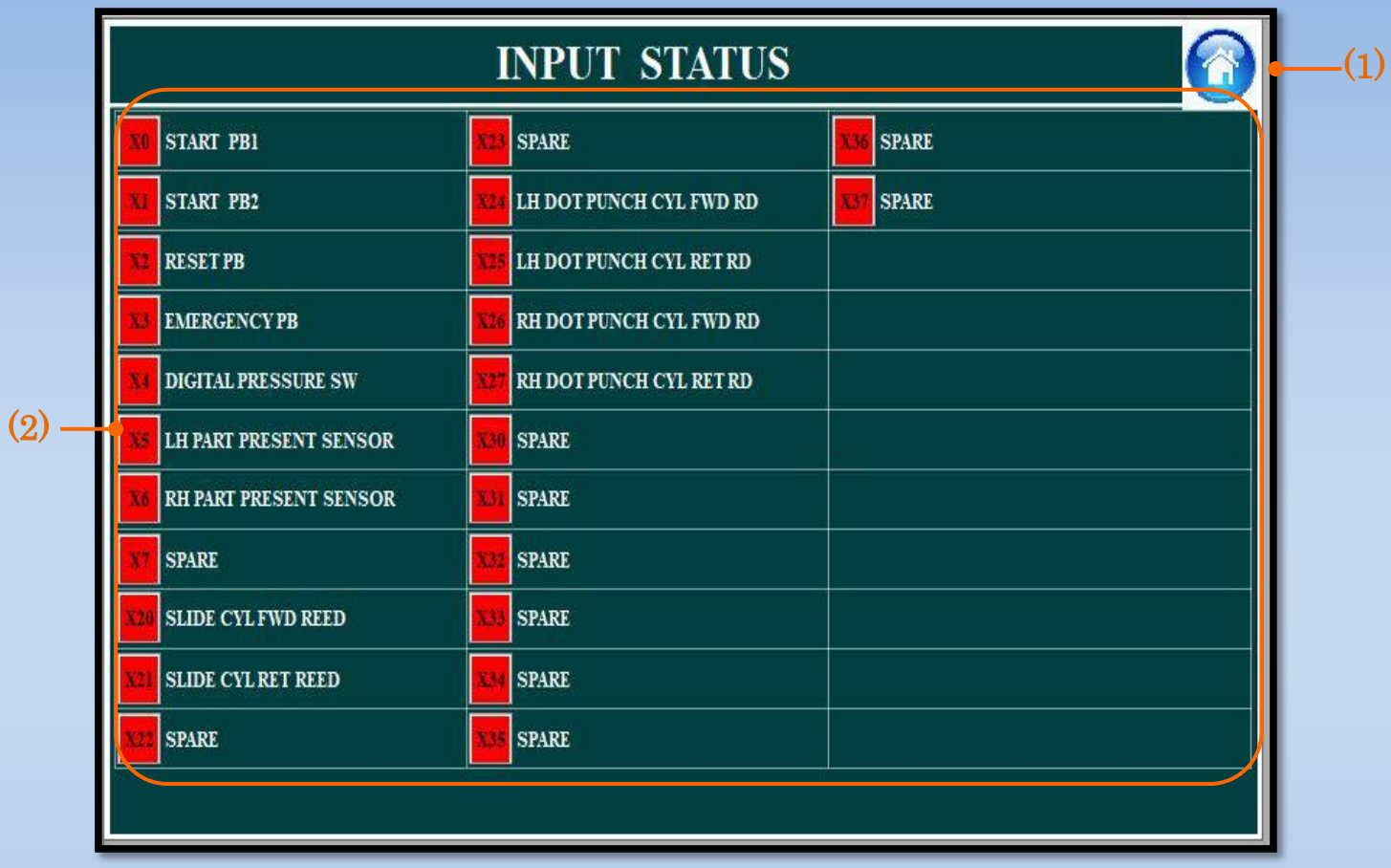
8. Part present sensor select/unselect setting option.

PROGRAM SELECTION SCREEN



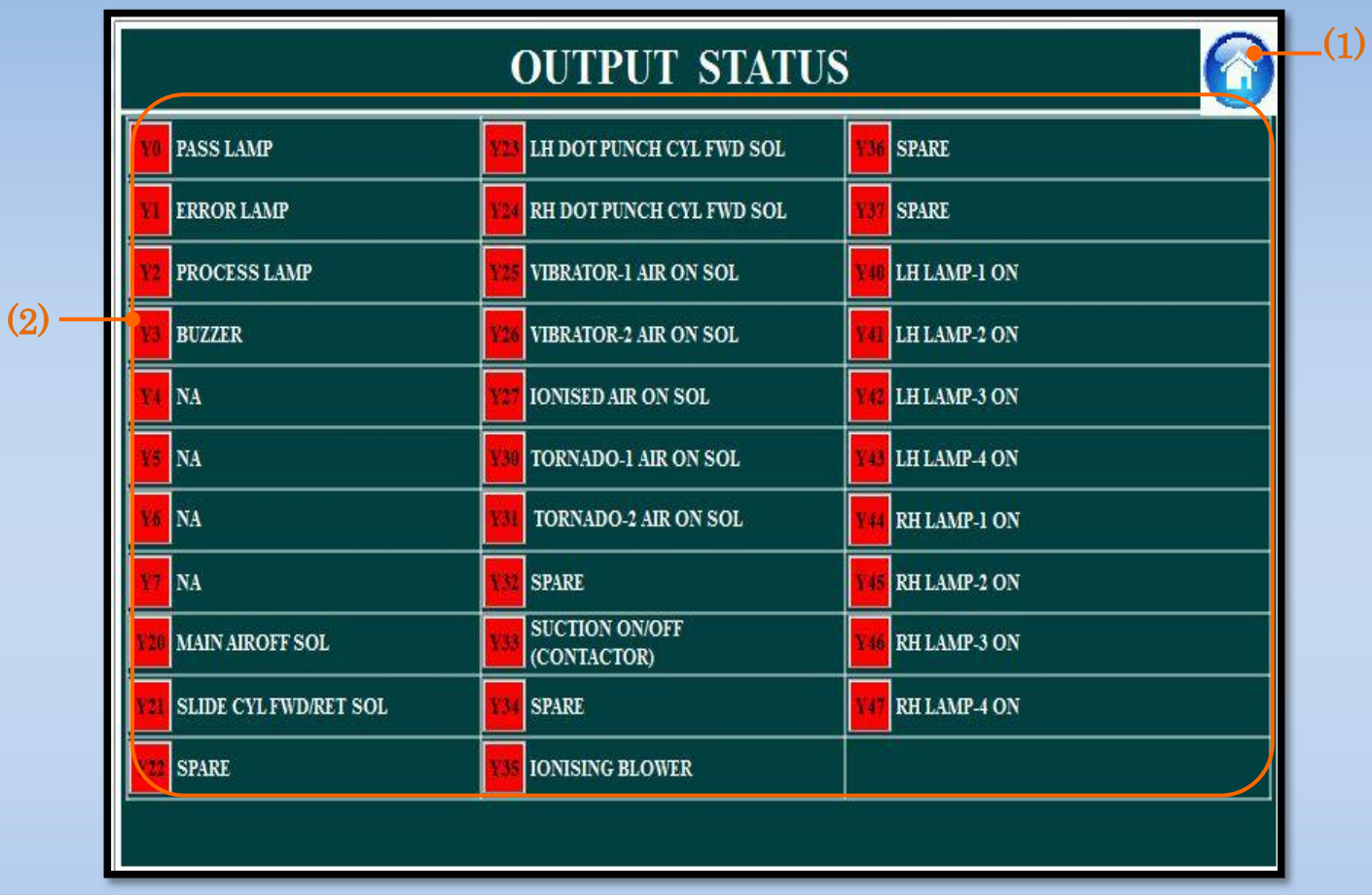
1. Home screen navigation button.
2. Program list.
3. Program number display.
4. Part count reset option.

INPUT STATUS



1. Home screen navigation button.
2. Plc input status.

OUTPUT STATUS



OUTPUT STATUS

Y0 PASS LAMP	Y23 LH DOT PUNCH CYL FWD SOL	Y36 SPARE
Y1 ERROR LAMP	Y24 RH DOT PUNCH CYL FWD SOL	Y37 SPARE
Y2 PROCESS LAMP	Y25 VIBRATOR-1 AIR ON SOL	Y40 LH LAMP-1 ON
Y3 BUZZER	Y26 VIBRATOR-2 AIR ON SOL	Y41 LH LAMP-2 ON
Y4 NA	Y27 IONISED AIR ON SOL	Y42 LH LAMP-3 ON
Y5 NA	Y30 TORNADO-1 AIR ON SOL	Y43 LH LAMP-4 ON
Y6 NA	Y31 TORNADO-2 AIR ON SOL	Y44 RH LAMP-1 ON
Y7 NA	Y32 SPARE	Y45 RH LAMP-2 ON
Y20 MAIN AIROFF SOL	Y33 SUCTION ON/OFF (CONTACTOR)	Y46 RH LAMP-3 ON
Y21 SLIDE CYL FWD/RET SOL	Y34 SPARE	Y47 RH LAMP-4 ON
Y22 SPARE	Y35 IONISING BLOWER	

1. Home screen navigation button.

2. Plc output status.

LH RH SELECTION SCREEN

1. Home screen navigation button.
2. Only LH side select button.
3. Only RH side select button.
4. LH & RH both side select button.

DO'S & DONT'S:

Every effort has been taken to ensure user safety while designing this machine. However, you are requested to adhere to following instructions to avoid any mishap.

- The machine should not be used for any other purpose other than that specified in the intended use.
- Do not operate the machine without understanding the installation, operation and maintenance manual thoroughly.
- Only trained, skilled, qualified and authorized personnel should install, operate and maintain the machine.
- Do not commission the machine without the aid of the qualified instructor.
- Do not reach into moving machinery.
- Never perform complex maintenance procedure single handed. Especially during start-ups as it is necessary to have another person manning the emergency stop in case of any problem.
- No tools other than that are prescribed should be used for machine maintenance.
- Do not store any loose tools in the machine compartments or control panel.
- If the machine is down for maintenance, place a sign over the control panel to inform others about that.
- Switch off the machine while connecting/disconnecting indo electrical connector.

CONTACT DETAILS

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